



Characterization of Organic Matter Composition of Soil and Flow Path Surfaces Based on Physicochemical Principles—A Review

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Abstract

Surfaces of macropores in structured soils can be made of clay-organic coatings on soil aggregates or linings on worm burrow or root channel walls. The outermost layer of such surfaces is mostly covered by organic matter (OM), which finally controls sorption properties that are relevant for preferential flow and transport. However, the OM properties and local distributions along flow path surfaces are largely unknown, and methods for analyzing intact surfaces are limited. This chapter reviews the principles and applications of Fourier Transform infrared (FTIR) spectroscopy in comparison with complementary nuclear magnetic resonance technique; it describes FTIR techniques for analyzing the composition of bulk soil OM and of OM fractions and of the millimeter-scale spatial distribution at intact surfaces in structured soils for comparing OM composition with cation exchange capacity of OM and soil wettability. Maps of the millimeter-scale heterogeneous spatial distribution of OM composition at aggregate and burrow surfaces could be obtained with diffuse reflectance FTIR (DRIFT) mapping technique. The distribution of OM composition suggests that sorption properties of the OM and wettability of macropore surfaces are also spatially variable at this local scale. DRIFT mapping technique requires relatively smooth and fine-textured intact sample surfaces; further developments need to account for effects of microtopography on the scattering of infrared light. The results indicate yet unknown implications for preferential flow and transport in structured soil, especially for reactive solutes.



1. INTRODUCTION

The global carbon (C) content within organic matter (OM) of mineral soils is estimated to be about 1550 pg in the upper 100 cm layer (e.g., [Bavaye et al., 2011](#); [Powelson, 2005](#)), which is about two times the atmospheric C content (e.g., [Baveye, 2007](#)). This amount does not consider organic C (C_{org}) stored in permafrost soils (i.e., 450 pg, [Zimov et al., 2006](#)) and peats (i.e., 165 pg, [Holmen, 1992](#)). Thus, soil organic matter (SOM) is an important C pool in the global C cycle. The main exchange fluxes for SOM, biomass production and SOM decomposition, are represented by the CO_2 exchange between the soil and the atmosphere (e.g., [Kuzakov, 2011](#)). The soil CO_2 sequestration depends on soil functioning as a sink or as a source (e.g., [Gregorich et al., 2005](#)). The important contribution of microbial SOM decomposition is strongly depending on moisture and temperature (e.g., [Conant et al., 2012](#); [Dungait et al., 2012](#); [von Luetzow and Koegel-Knabner, 2009](#)).

However, fluxes of dissolved organic carbon (DOC) in soil solution toward ground and surface waters (e.g., [Mei et al., 2012](#); [Wagai and](#)

Sollins, 2002) and fluxes of CO₂ in the gas phase (e.g., Gregorich et al., 2005; Xu et al., 2008) toward the atmosphere are normally relatively small in relation to the soil C pool (i.e., the SOM content). Smith et al. (2008) found that the rates of soil organic carbon (SOC) stock changes in response to global warming were highly uncertain because of relatively small net changes in soil C-pools as compared to mostly larger biomass and litter C stocks. For improved understanding, Kuzyakov (2011) suggests linking the investigations of pools and fluxes.

Irrespective of global C cycling problems, the SOM content has been used as an indicator for characterizing and understanding soil management- and land use-induced effects on cropping and soil quality (e.g., Hayes and Clapp, 2001; Stevenson, 1994). The term “soil quality” has been defined (e.g., Karlen et al., 1997) as “. . . the capacity of the soil to perform ecological functions. . . .” The role of SOM in maintaining soil quality has been emphasized by many authors (e.g., Powlson, 1996; Reeves, 1997; Smith et al., 2008; Sposito, 2008; Stevenson, 1994).

The quality of SOM itself depends on the chemical composition (e.g., Gärdenäs et al., 2011; Kögel-Knabner, 2000; Leifeld and Kögel-Knabner, 2005). Differently soluble SOM fractions (or otherwise physically and chemically separated OM fractions) have been used for characterizing management-induced changes in the SOM composition with direct effect on soil properties (e.g., Ding et al., 2006; He et al., 2009; Munoz Garcia and Faz Cano, 2012; Sequeira and Alley, 2011; Simon et al., 2009). Instead of using the methodologically defined SOM fractions, a model for continuous SOM quality distributions was proposed (Bruun et al., 2010) that characterized all the different SOM particles and fractions with respect to density, particle size, or other properties.

The properties of coarse-textured soils are assumed to be more affected by changes in SOM than those of loamy soils (Stevenson, 1985). While the soil texture practically remains the same, the SOM can be affected by land use and soil management with respect to both the content and the composition (Jenkinson, 1988). The SOM has been reported to affect several important soil functions (Van Lauwe and Giller, 2006) such as aggregate stability and storage capacity for water and nutrients (Baldock and Nelson, 2000), the cation exchange capacity (CEC) (e.g., Celi et al., 1997), and the wettability (Capriel, 1997; Chenu et al., 2000). However, relations between SOC content and soil properties could only in certain cases be explained simply by the SOC content (e.g., Horne and McIntosh, 2000). Negre et al. (2002) suggested considering the macromolecular behavior

of SOM that depends on chemical composition and three-dimensional spatial structure of the SOM to explain soil wettability, although the detailed relations are largely unknown (e.g., [Bachmann and McHale, 2009](#); [Ellerbrock et al., 2005](#)). The chemical composition of the SOM was found to also control its CEC (e.g., [Almendros, 1995](#); [Gressel et al., 1995](#)). In analogy to defined OM ([Günzler and Böck, 1990](#)), the negatively charged carboxyl groups in SOM are mainly determining the CEC. Here, the term “SOM composition” describes the type and quantity of functional groups (e.g., carboxyl, alkyl, or hydroxyl groups) and their conformation describing the spatial arrangement at the molecular scale ([Morrison and Boyd, 1983](#)). The SOM composition therefore characterizes the functional properties of OM in soil ([Stevenson, 1985](#)). Soil type, climatic conditions, land use, crop rotation, and soil management modify the SOM composition indirectly through effects on the water and element balances of the ecosystem (e.g., [Antil et al., 2005](#); [Kaiser and Ellerbrock, 2005](#); [Kaiser et al., 2007](#); [von Lützow et al., 2006](#)); the input from crop and root residues ([Francioso et al., 2000](#)) and the processes of SOM decomposition ([Mathers and Xu, 2003](#)). Such effects on SOM composition can preferably be studied using soils of long-term field experiments (LTFE) (e.g., [Quideau et al., 2001](#)) since changes in land use and management will take many years of constant management before a new equilibrium is reached (e.g., [Schmidt et al., 2000](#); [West and Post, 2002](#)).

For analyzing OM functional groups, several spectroscopic techniques have been applied, such as infrared (IR), Fourier Transform infrared (FTIR) (e.g., [Capriel et al., 1995](#); [Demyan et al., 2012](#); [Gerzabek et al., 1997](#)) or nuclear magnetic resonance (NMR) spectroscopy (e.g., [Hayes and Clapp, 2001](#); [Kogel-Knabner, 1997](#)). In the field of organic chemistry, such spectroscopic techniques were used to determine the specific molecular structure of pure, exactly defined organic substances (e.g., [Hesse et al., 2005](#)). These techniques are based on the interactions between electromagnetic radiation and the molecular subcomponents (e.g., electrons, magnetic spin of certain isotopes, or molecular bonds) within the functional groups. During the wave-particle interaction, the light transfers a part of its energy on the molecular subcomponents (e.g., [Hesse et al., 2005](#)). Such energy transfer will only take place if ([Staab, 1964](#)) the energy of the light is proportional to the differences in the energy of the molecular component in the excited minus the basic energy levels (i.e., quantization of energy levels). The energy level of such subcomponents depends on the type of functional groups within the molecule and the possible inter- (i.e., between different

molecules) and intra (i.e., within the same molecule) molecular interactions such as hydrogen bonding or complex formation with polyvalent cations (PC) such as Ca^{2+} , Al^{x+} , Mn^{x+} , or Fe^{x+} .

The intensity of the light of a defined wavelength remains unchanged if certain functional groups are absent. In addition, the interaction between a C=O group and a cation can cause, for example, shifts in the absorption bands of the C=O group in both FTIR and NMR spectra. The information from NMR and FTIR is basically the same, however, due to the different wavelengths; NMR analysis of molecular components is based on the reversal of the orientation of the magnetic spin of certain isotopes, while FTIR is based on the excitation of vibration and rotation of molecular bonds. This overview demonstrates that the spectral techniques have specific challenges and limitation that help to identify and to distinguish SOM functional groups. [MacCarthy and Rice \(1985\)](#) as well as [Simpson et al. \(1997\)](#), among others, used spectral techniques to determine molecular structures for humic acid fractions. In contrast to a defined organic substance, the heterogeneous mixture of operationally defined fractions in SOM is characterized by differing transformation status, degradability, and turnover time ([von Lützow et al., 2006](#)) ranging from labile (e.g., microbial biomass) to relatively inert components ([Baldock and Nelson, 2000](#); [Stevenson, 1994](#)).

Considering its complex nature, [Stevenson \(1982\)](#) suggested a simplified schematic molecular structure model for SOM that was consisting of different functional groups such as alkyl, carboxyl, and hydroxyl groups ([Fig. 3.1](#)). The spectra obtained from analyzing such complex organic substances thus integrate over the spectral properties of all components present in the sample reflecting the OM functional group composition of the mixture.

Note that spectroscopic techniques always provide molecular-level information on the composition of the samples. For analyzing, for instance, the behavior of solutes in a soil profile, upscaling and transformation of molecular information to profile or landscapes scales will become necessary. The use of spectral information involves transformation in effective transport properties (i.e., sorption coefficients) and the consideration of spatial characteristics at the profile scale (e.g., [Carter, 2000](#); [Müller et al., 2007](#)); the soil spatial variability in SOM composition is additionally important at the field (e.g., [Farenhorst, 2006](#); [Sommer, 2006](#)) and larger scales (e.g., [Prechtel et al., 2009](#); [Viaud et al., 2010](#)). For such transformation, it becomes necessary to consider sample preparation and sampling technique, the spatial distribution of SOM composition in soil profile as well as at structural

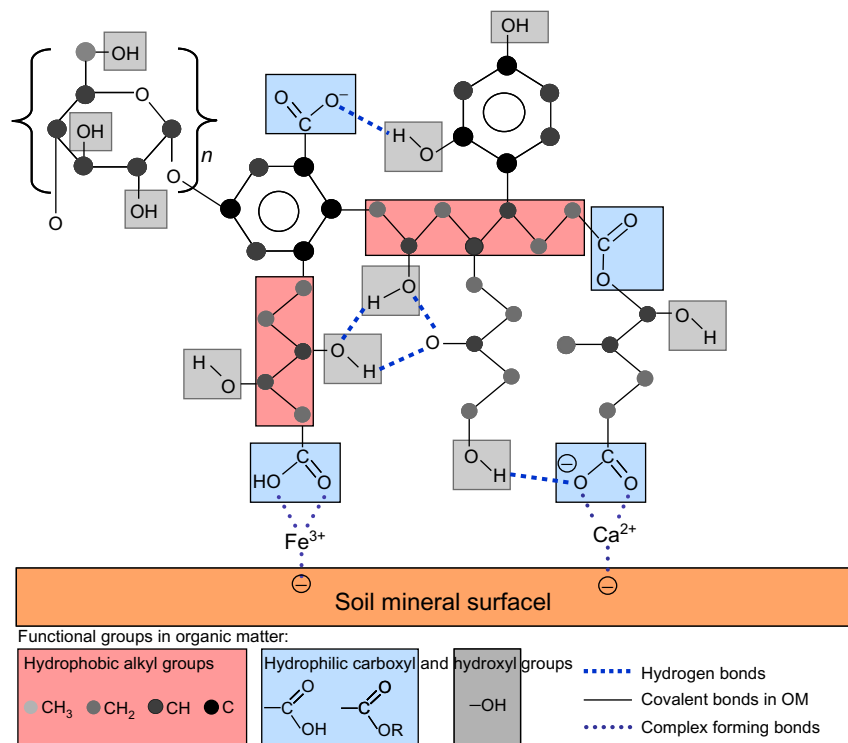


Figure 3.1 A schematic structure for molecules in soil organic matter consisting of alkyl (reddish), carboxyl (bluish), and hydroxyl (no color) groups as well as aromatic rings. The negatively charged mineral surface is complexing the organic molecular structures by polyvalent metal cations such as Ca^{2+} , Al^{x+} , Mn^{x+} , or Fe^{x+} . The bottom scheme is indicating the alkyl groups different in hydrogen saturation; the carboxyl groups that can have OH or OR; R denotes the alkyl chain. Adapted from [Stevenson \(1982\)](#).

surfaces together with above-mentioned challenges and limitations of the spectroscopic techniques.

In soils, the CEC and chemical sorption reactivity for dissolved substances generally increase with SOM content ([Sposito, 2008](#); [Stevenson, 1994](#)), while the wettability of soil decreases with SOM content in many cases (e.g., [Chenu et al., 2000](#); [Jaramillo et al., 2000](#); [Mataix-Solera and Doerr, 2004](#)) with some exceptions explained by differences in the composition of SOM (e.g., [Horne and McIntosh, 2000](#)). Such results are for (i) homogeneous soils, (ii) assuming water percolation in a uniform wetting or moisture front with (iii) all available pores and particle surfaces uniformly wetted, and (iv) in contact with the soil solution.

In structured soils, preferential flow may occur along earthworm burrows, root channels, or the interaggregate pore space (e.g., Beven and Germann, 1982; Flury et al., 1994; Ghodrati and Jury, 1992; Nieber, 2000). When moving through such pathways, the percolating water is bypassing much of the volume of the porous soil matrix (for details, see reviews of, e.g., Jarvis, 2007; Köhne et al., 2009a). The rapid movement through small fractions of the soil volume strongly limits the soil's filter and buffer function (e.g., Köhne et al., 2009b). Moreover, any contact and exchange between the preferentially moving water and the surrounding soil matrix is largely restricted to the surface areas of such flow paths. Despite the increasing progress in describing preferential flow processes (e.g., Gärdenäs et al., 2006; Ray et al., 2004; Simunek and van Genuchten, 2008; Simunek et al., 2003), predictions of such processes are limited (Clothier et al., 2008) because it is difficult to determine the soil structural parameters *in situ* (Köhne et al., 2009b). Additionally, the functional properties (e.g., wettability) of surfaces at preferential flow path are hard to analyze; up to now they were quantified for mixed soil samples or for an aggregate packing (Goebel et al., 2008; Ramírez-Flores et al., 2008).

As for mineral particles, the surfaces of porous soil aggregates are mostly covered by OM. Thus, the OM composition at these surfaces may finally control wettability and sorption and transfer properties of flow pathways. The assumption that the spatial variability of water transfer is caused by local differences in wettability of flow path surfaces (Hallett et al., 2004) could be shown by staining experiments (Kodesova et al., 2012; Lipsius and Mooney, 2006). The small-scale local heterogeneity in SOM distribution seems to be of special importance for pesticide transport and chemical sorption processes in structured soils (e.g., Jury and Flühler, 1992; Kaiser and Zech, 1998; White, 1985). While surfaces of aggregates and pores are intensively studied for the mineralogical properties (e.g., Celis et al., 1997; Jongmans et al., 1998; Kaiser and Wilcke, 1996), there are only a few studies on the effect of SOM composition on preferential flow. When analyzing SOM, we need to consider that the spatial distribution of OM in soil is locally highly variable with consequences for turnover rate processes and transport properties (Fig. 3.2A; Leue et al., 2010).

Such small-scale heterogeneity could not be considered when analyzing mixed soil samples (Fig. 3.2B) because the spatial distribution of SOM at preferential flow paths will be lost when mixing the sample. In addition, the spatial geometry of the patchy-like distribution of SOM at structural surfaces is lost. The transport in pathways is local; therefore, the local distribution of both the solution and the solid phase has to be considered for

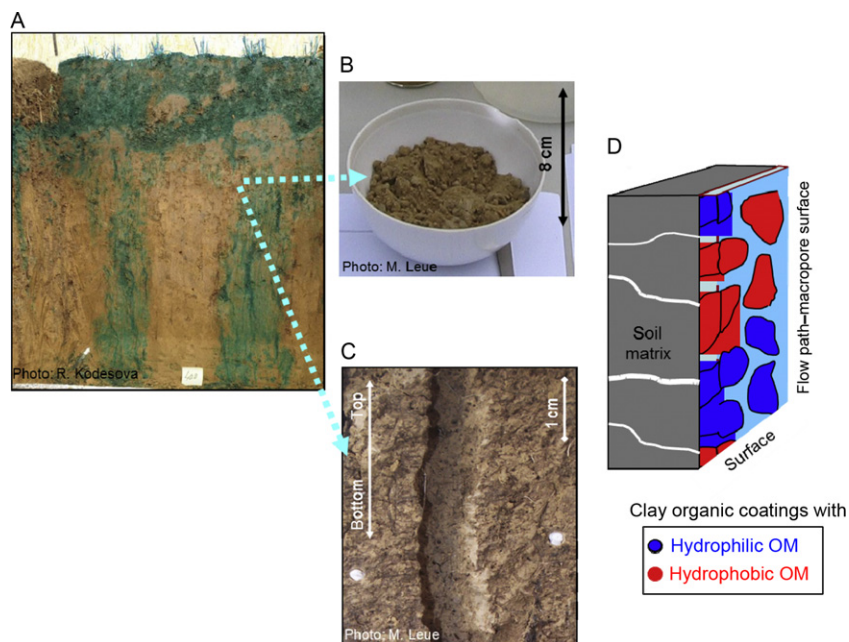


Figure 3.2 Photos of (A) a Luvisol soil profile, (B) a mixed and (C) an intact structural surface sample from a structured soil, and (D) a scheme of the hypothesized spatial distribution of SOM at soil structural surfaces. The Luvisol soil profile is located at the site near Hnevceves, Czech Republic, Brilliant Blue tracer experiment, R. Kodesova, University of Prague.

predictions. Thus, for analyzing the small-scale distribution of SOM at structural surfaces (Fig. 3.2C), it is necessary to investigate intact samples.

Swift (1999) reviewed, for instance, different wet chemical and spectroscopic procedures (FTIR and NMR) for analyzing SOM in bulk soil samples and SOM fractions. However, such investigations were mostly done by different working groups by using either NMR or FTIR except for Inbar et al. (1989) which compared the physical principles of NMR and FTIR for analyzing the same OM fractions from compost samples. The application of the spectroscopic techniques for intact surfaces was yet presented neither in that detail nor in relation to soil physical and chemical properties.

The objective of this review is to discuss the basics of complementary spectroscopic techniques like NMR and FTIR to describe challenges and limitations of spectroscopic techniques and to distinguish between problems caused by soil sample properties, sampling procedure, and basic principles of spectral technique; to describe the application of FTIR techniques for

analyzing composition of (i) defined pure organic compounds, (ii) SOM of mixed soil samples, and (iii) for exploring the small-scale spatial variability of SOM composition at intact structural surfaces (preferential flow path).

The information obtained by FTIR spectroscopy may be related to other soil properties and local two-dimensional (2D) distributions of SOM composition compared with water infiltration to indicate possible use of spectral information for describing SOM properties that are relevant for transformation and transport processes in structured soil.



2. SOM QUANTIFICATION AS SOC

The SOM content makes up about 1–6% of the bulk soil mass of many mineral arable and grassland soils (Stevenson, 1994). Although the fraction is relatively small, SOM is an important component controlling soil physical, chemical, and biological properties. Information on SOM content is usually included in standard soil characterizations together with texture data, among others. The SOM content of soil samples is determined as the content of organic C (C_{org}). However, C_{org} is just one component of SOM besides other elements such as N, S, P, and varying cations in SOM (Sposito, 2008). For converting in general SOM characterizations the measured C_{org} contents to the SOM contents, a relation has been proposed and a factor of about 1.72 (e.g., Nelson and Sommers, 1996) between C_{org} and SOM used. Considering the highly heterogeneous composition of SOM, it is clear that this relation is only an estimate that depends strongly on soil genesis and land use among other factors (e.g., Nelson and Sommers, 1996). Despite these limitations, the widely available C_{org} data have been used for discussing effects of environmental, management, and site factors on amount of SOM as follows: Jenny et al. (1949), for example, compare nitrogen and OM production and decomposition in temperate and tropical soils. The data of this study have been used in more than 400 articles to explain results on increasing organic carbon (OC) contents or litter decomposition among other aspects (Clapp and Hayes, 2006). In a similar way, the data of Jenkinson and Rayner (1977) were used until today. The C_{org} contents were found to depend on climate, soil texture, land use (e.g., West and Post, 2002), crop rotation, and fertilization (Gregorich et al., 2005). The changes in the C_{org} contents of soils could occur on relatively short (Leinweber et al., 1994) and long time intervals (Adani and Tambone, 2005). Most short-term changes in C_{org} contents were possibly restricted to the decomposable OM fractions (e.g., Haynes, 2000, 2005).

Analytically, the C_{org} content is mostly determined (e.g., [Nelson and Sommers, 1996](#); [Sposito, 2008](#)) from the difference between total (C_t) and carbonate carbon (C_{CO_3}) as:

$$C_{\text{org}} = C_t - C_{\text{CO}_3} \quad (3.1)$$

The content of C_t can be determined after dry combustion at 1250 °C ([DIN ISO 10694, 1994](#)) by IR detection of CO_2 (e.g., CNS 2000, LECO Ltd.) released from the soil sample. The content of C_{CO_3} has mostly been determined after application of phosphoric acid by gas-chromatographic analysis of carbon dioxide (CO_2) evolution ([Nelson and Sommers, 1996](#)); an overview of other methods can be found, for instance, by [Chatterjee et al. \(2009\)](#).

Examples for the change in C_{org} contents have been reported from soils of LTFE located in Bad Lauchstädt, Halle, Müncheberg (Germany) and Rothamstedt (UK). For the 100-year-old fertilization experiment located at Bad Lauchstädt, the C_{org} content of the loamy silt soils was found to be significantly correlated with management practices and crop yields ([Körschens and Müller, 1996](#)). Similar relations were observed for the C_{org} contents of the soils in Rothamstedt (e.g., [Powlson, 1996](#)). In contrast for the LTFE experiment in Müncheberg located at coarse-textured sandy soils, no significant changes in the C_{org} contents for different fertilizer applications were observed ([Ellerbrock et al., 1999a,b](#)). The changes in the C_{org} content of the total soil in response to alterations in land use or management practice are difficult to detect because of variability and heterogeneity.

The C_{org} content changes are mostly small as compared to the generally high background levels and spatial variability in C_{org} contents ([Haynes and Beare, 1996](#); [Powlson, 1996](#)). However, when focusing on certain (i.e., preferably labile) C fractions, the changes in these fractional C-contents could be better related to land use or soil management practices and observed in shorter time periods as compared to analyzing the total soil C_{org} contents (e.g., [Haynes, 2005](#)). For example, the hot water extracts (e.g., [Körschens et al., 1990](#)) were significantly varying between fertilizer applications at the LTFE in Bad Lauchstädt, Thyrow, and Müncheberg and more than for C_{org} total.



3. SOM SEPARATION METHODS

SOM can be separated by physical and chemical procedures ([Blazejewski et al., 2005](#); [Hayes, 2006](#); [Stewart et al., 2008](#)) into fractions

different in composition and stability (e.g., [Olk and Gregorich, 2006](#)). The SOM fractions can generally be separated into particulate and soluble OM fractions. The term particulate organic matter (POM) has mostly been used to describe a density fraction obtained from particle-size fractions; the term light fraction (LF) defines a density fraction obtained by similar procedures but from the bulk soil ([Gregorich et al., 2006](#)).

The SOM fractions have mostly been separated from mixed samples of soil horizons (e.g., [von Lützow et al., 2007](#)), while studies focusing more on the process–structure interactions used spatially defined sampling (e.g., [Ellerbrock and Gerke, 2004](#)) or even noncontact sensing methods on intact samples ([Ellerbrock et al., 2009](#)). The procedures and resulting fractions and the possibilities to characterize SOM composition are summarized in [Fig. 3.3](#). From mixed samples ([Fig. 3.3B](#)), the POM, LF, and soluble SOM fractions are characterized as a mean composition integrated over the bulk sample volume. From the spatially defined samples ([Fig. 3.3A](#)),

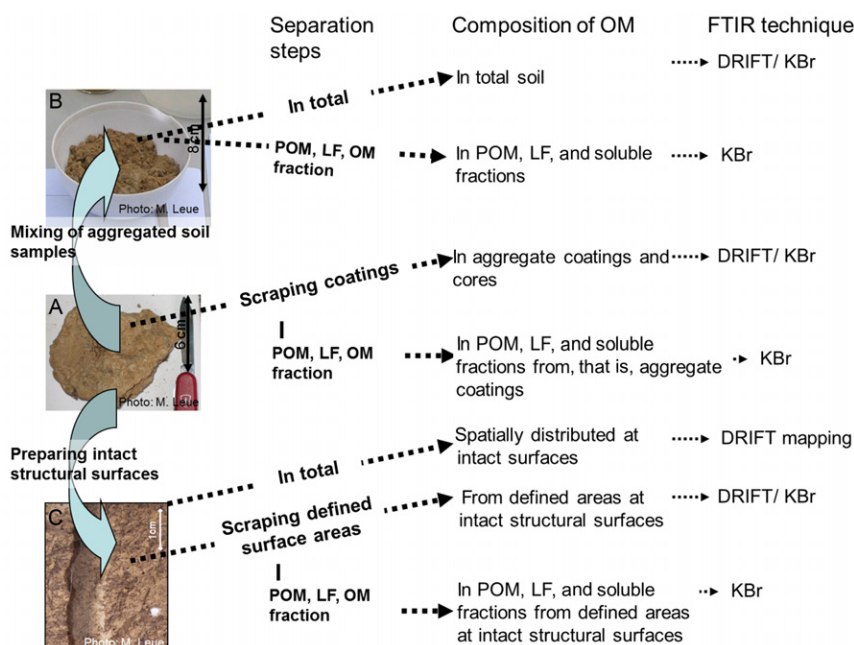


Figure 3.3 Sample preparation steps of aggregated soil: (A) intact structural surface, (B) mixing, and (C) peeling or scraping of coatings in total or from defined surface areas, and procedures for separating POM, LF, and soluble OM fractions from mixed and intact soil samples. The FTIR spectroscopic techniques for analyzing the OM of the different sample types refer to sections 3.1 to 3.4.

the same SOM fractions are characterized only for the sampled regions (e.g., of aggregate coatings). From intact samples (Fig. 3.3C), only the total SOM can be characterized, reflecting a mean composition of all fractions at the sensed surface. The SOM fractions from the various sampling and separation procedures can be used for subsequent spectral analysis (Fig. 3.3), either FTIR or DRIFT to characterize the composition of SOM fractions. One advantage of using SOM fractions for SOM characterization is that the impact of soil mineral components on the spectra is relatively small as compared to using bulk soil (Janik et al., 1995).

In one of the first approaches (Kononova, 1958), soluble SOM fractions were separated from soil by using a 0.1 M sodium hydroxide (NaOH) solution: The resulting solution was then treated with 2 M hydrogen chloride (HCl) to obtain

- i. a NaOH-soluble fraction that is insoluble in HCL called *humic acid*,
- ii. a NaOH- and HCL-soluble fraction called *fulvic acid*, and
- iii. a residue that remains after NaOH treatment called *humins*.

This standard OM extraction with NaOH (e.g., Swift, 1996) has been criticized to strongly alter OM chemical characteristics (e.g., Hayes, 2006) because of the alkaline conditions. At pH values between 12 and 13, for example, saccharides may interact via Maillard reaction with protein components to result in melanoids (e.g., Ziehmman, 1996) and esters can be hydrolyzed, which results in organic acids and alcohols (e.g., Morrison and Boyd, 1983). To reduce possible effects of such alterations, alternative procedures for SOM separation have more recently been developed, which could be grouped into the three basic procedures: (i) density fractionation, (ii) sequential extraction, (iii) sequential oxidation, and (iv) a combination of density and fractionation and sequential extraction.

3.1. Density fractionation

The density fractionation separates POM fractions from soil and soil aggregates. For the physically unprotected POM fractions, heavy liquids of densities between 1 and 2.3 g cm⁻³ have often been used to separate POM from soil and particle-size fractions (e.g., Swift, 1996). Such heavy liquids separate easily decomposable POM from soil samples (Helfrich et al., 2007). The density fractionation procedures are often combined with the supply of dispersing energy (e.g., shaking with glass beads or ultrasonic treatment) (Oorts et al., 2005; Wander, 2004). Ultrasonic treatment, for instance, has been used to distinguish the LF occluded in macro- and microaggregates from the nonoccluded ones (e.g., Swift, 1996).

The POM and LF contents vary in a wide range for the different soils. For arable soils, the LF (i.e., the POM from the particle-size fractions) was found to represent about 1–5% of SOC in mineral topsoil horizons (e.g., [Ludwig et al., 2003](#); [Sohi et al., 2001](#)). The content of POM separated from arable soils represented an average of 19% (up to 65%) of SOC, whereas the LF fractions accounted for an average of 6% (up to 36%) while in forest soils, POM was found to represent about 27% in the mean (up to 48%) and LF about 8% (up to 74%) of SOC ([Gregorich et al., 2006](#)). The contents of a POM fraction, here separated by electrostatic attraction, were mostly small; values of up to 25% of SOC were found for one forest soil ([Kaiser et al., 2009](#)). The POM and LF fractions from mineral topsoil layers were reported to be sensitive to climatic, land use, or management changes because of their relatively fast turnover and close connection to litter input ([Bird et al., 2002, 2007](#)). Some authors found that the decomposition rate of POM decreased with size of the sieved fraction where the POM has been separated from [Guggenberger et al. \(1994a,b\)](#). Such relations were used, for instance, in C-turnover models to describe an increase in the mean residence time of differently stable C-pools ([Six et al., 1998](#)). The nonoccluded LF was found more easily decomposable by microbes than occluded LF ([Roscoe et al., 2004](#)). Charred organic material (i.e., black carbon) in POM ([Quénée et al., 2006](#)) was made responsible for an increase in the mean residence time determined by ^{14}C dating ([Baisden et al., 2002](#); [Crow et al., 2007](#)). However, density separation needs to be combined with isotope tracers and detailed chemical characterization as well as other separation techniques for improving models that link SOM dynamics with POM and LF fractions (e.g., [Wagai et al., 2009](#)).

Problems associated with fractionation by using heavy liquids such as sodium polytungstate (NaPT) and sodium iodide solutions ([Wander, 2004](#)) are caused by (i) triggering solubilization, (ii) altering chemical characteristics, and (iii) promoting C transfers between organic particles and soluble OM fractions ([Magid et al., 1996](#)). The separation of organic particles using a NaPT solution of a density of 1.6 g cm^{-3} resulted in organic C losses of 17–26% ([Crow et al., 2007](#)). Besides losses through dissolution processes, the dispersions of organo-mineral compounds could affect the separation of organic plant residues or mineral-associated OM ([Kaiser and Guggenberger, 2007](#)).

3.2. Sequential extraction procedures

Independent of density and size, SOM can be divided into differently soluble fractions. A number of studies found that their solubility strongly depended

on interactions of SOM with PC and mineral surfaces (e.g., Baldock and Skjemstad, 2000; Kleber et al., 2004; Mikutta et al., 2005b; Sposito, 2008; Stevenson, 1994; Wattel-Koekkoek et al., 2003). Interactions of SOM with PC reduced the solubility of OM in soil (Sposito, 2008). Furthermore, the mineral-associated SOM was frequently assumed to be protected against microbial decay and should therefore be more stable than the water-soluble fractions (e.g., Baldock and Skjemstad, 2000; Gillet and Ponge, 2002; Gregorich et al., 2003; MacCarthy et al., 1990). In contrast, the hot- (Körschens et al., 1990) and cold water- (e.g., Kalbitz et al., 2003) soluble SOM fractions were found more easily decomposable. An acid hydrolysis was proposed for separating more easily decomposable OM fractions such as sugars (Paul et al., 1997). A sodium pyrophosphate (NaPY) solution (Bremner and Lees, 1949) was assumed to yield OM fractions (NaPY-soluble OM) that are stabilized by interactions with PC and clay minerals (Wattel-Koekkoek et al., 2003), oxides and hydroxides (Kaiser and Zech, 1996), and metal-organic complexes (Masiello et al., 2004). During longer extractions (≥ 24 h), the composition of soil minerals may be affected by the high pH value (i.e., pH of 10) of the NaPY solution (Kaiser and Zech, 1996).

For analyzing differently soluble SOM fractions from the same sample, a sequential extraction procedure was recently proposed by Ellerbrock and Kaiser (2005) combining the water with the NaPY extraction. In a first step, the water-soluble SOM fraction was removed from the sample (water-soluble OM), and in a second step, the sodium pyrophosphate-soluble fraction (NaPYseq-soluble OM) was separated. The OM in the solid residue after the second step was assumed to represent the slowest decomposable fraction obtained by this procedure. The amount and the composition of the sequentially extracted SOM fractions are specific for soil mineral and texture. The water and NaPYseq-soluble fractions amount around 30% up to 50% of SOC (Kaiser and Ellerbrock, 2005; Kaiser et al., 2007). The composition of the sequentially extracted SOM fractions was found useful to distinguish SOM of different soil management and land use (Ellerbrock et al., 1999b, 2001a; Kaiser et al., 2007).

3.3. Sequential oxidation

This approach uses oxidizing substances for a stepwise chemical decomposition of SOM (e.g., Plante et al., 2004). The SOM fractions obtained by the sequential oxidation procedure are assumed to reflect differences in microbial decomposability (Balesdent, 1996); the more rapidly oxidizing fractions are assumed to be more easily decomposable. The CO_2 released during the

oxidation procedure allowed determining OC contents of these SOM fractions (Mikutta et al., 2005a). However, in contrast to SOM fractions obtained by sequential extraction and density fractionation, a spectral characterization of the oxidized SOM fractions is not feasible.

3.4. Combination of density separation and sequential extraction

The combination of the two approaches, density fractionation and sequential extraction, separates the particulate from the soluble OM in a stepwise procedure (e.g., Kaiser et al., 2009). The removal of POM prior to the extraction prevents from an unintended mixing of soluble components in the POM with those of the remaining SOM. To avoid losses of soluble components when separating root residues (i.e., analogy to POM and LF) from soil, Arinushkina (1970, p. 111) applied an electrostatic attraction procedure instead of a density fractionation. Kuzyakov et al. (2001) could separate between 43% for sandy and loamy soils and 35% for a silt loam of the added ^{14}C -labeled roots by using a modified electrostatic method. A preseparation of soil samples in sieved-size fractions improved distinction between organic and mineral particles attracted by electrostatic forces (Kaiser et al., 2009).

The example of the electrostatically separated relatively coarse POM fraction >0.315 mm according to Arinushkina (1970) from an air-dried forest topsoil (Fig. 3.4A) indicates that it consists mainly of fine roots, partly decomposed leaves, and wooden pieces that passed through the 2-mm sieve. The samples from arable topsoils obtained by density fractionation using

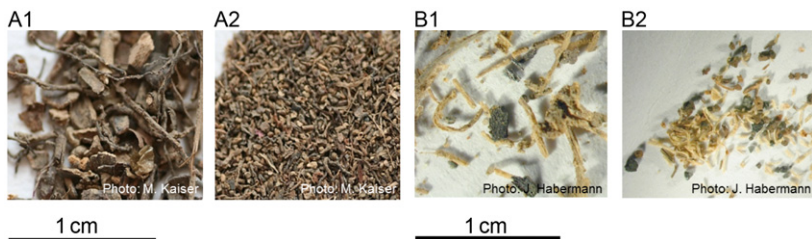


Figure 3.4 Photos of (A) particulate organic matter (POM) separated by electrostatic attraction (Kaiser et al., 2009) and (B) of the light fraction (LF) separated by a density of 1 g cm^{-3} (Kaiser et al., 2007). The particle-size fraction of the POM of the forest soil is 1.25–2 mm (A1) and 0.4–0.5 mm (A2). The LF is from the arable bulk soil of the long-term field experiment in Halle obtained before (B1) and the additionally obtained after (B2) ultrasonic treatments (i.e., showing smaller particles that were occluded in aggregates).

water (Fig. 3.4B) and ultrasonic treatment (Fig. 3.4C) consisted of fine roots as in Fig. 3.4A but no wooden chips. The decomposed leaves in Fig. 3.4C were smaller as those in Fig. 3.4A and B.



4. METHODS FOR DETERMINATION OF OM COMPOSITION

In chemistry, spectral techniques like NMR or FTIR spectroscopy are complementary methods to analyze the composition of *pure and defined* organic components (e.g., Hesse et al., 2005). In general, those techniques act like a question–answer procedure. The question was asked by electromagnetic radiation (light) different in its energy (caused by difference in wavelength). And the answer will be given by the sample: It either interacts with light of certain energy or not, resulting in absorption of light (which causes a decrease in light intensity) or not.

4.1. Physicochemical principles of the methods

The energy of the light (i.e., electromagnetic radiation) is defined by its frequency

$$E = h \cdot \nu \quad (3.2)$$

with E , the energy of the light (J), h , the Planck's action quantum (J s^{-1}), and ν , the frequency of the light (s^{-1}), which is defined by the speed of light, c (cm s^{-1}) divided by the wavelength of the light λ (cm)

$$\nu = \frac{c}{\lambda}. \quad (3.3)$$

All spectroscopic methods are based on the wave–particle dualism of the light (Hesse et al., 2005). Because of this dualism, light of a certain wavelength could give molecular subcomponents (e.g., atoms, protons, electrons) a push and a probable transfer of energy. During this “push,” a molecular subcomponent is shifted toward a higher energy level. Each of these subcomponents has in principle two defined energy levels: a ground level and an excited level. A transfer between ground and excited level takes place if the energy of the electromagnetic radiation is equal to the energy difference between excited and ground level (Hesse et al., 2005). The energy levels of the molecular subcomponents (e.g., magnetic spin of atoms) depend on the physicochemical properties of functional groups, which can be identified.

The transfer of electrons from ground to an excited energy level requires a light with a relatively high amount of energy (e.g., UV or X-ray). The energy content of light required causing vibration and rotation of molecular bonds is much lower (e.g., Raman spectroscopy, FTIR; Hesse et al., 2005). Each spectroscopic technique uses light of different energies affecting different subcomponents within the molecules:

- The IR light causes vibration or rotation of molecular bindings,
- UV–vis (i.e., the UV and the visible light) light causes a transfer of electrons from occupied into nonoccupied orbitals (Hesse et al., 2005), and
- NMR affects the orientation of nuclear spins within an external magnetic field by radio waves (Breitmaier and Voelter, 1989).

For OM molecules containing C=O groups, for example, the FTIR light is causing a vibration of the double bond between C and O, the UV–vis is causing a “jump” of an electron of the C=O double bond into a non-occupied orbital, and the radio waves in ^{13}C NMR are causing a change in the orientation of the nuclear spin of a ^{13}C atom in the C=O group. The spectroscopic methods are all based on the same principle; they differ in the wavelength they use and the molecular subcomponents they interact with (Staab, 1964). In all spectroscopic techniques, the intensity of the light absorbed by the sample is determined as a difference in light intensity. For each wavelength, the intensity of the light is registered by a detector yielding a spectrum of the intensities of the light of all wavelength after transfer through the sample.

4.2. NMR spectroscopy

NMR is based on the fact that certain isotopes of H (^1H) and C (^{13}C) show a magnetic moment and that the orientation of the corresponding atomic spin within an external magnetic field could be changed by the electromagnetic radiation of a certain wavelength (Hesse et al., 2005). The ^{13}C NMR spectroscopy focuses on the ^{13}C isotope. The magnetic property of a ^{13}C spin depends on the binding status of the corresponding C atom within the organic molecule (Breitmaier and Voelter, 1989). Based on that dependency, ^{13}C NMR spectroscopy allows differentiating the C atoms within a molecule according to their binding status (for determining the type of OM functional groups) because such differences will cause a difference in chemical shift (ppm) of peaks within the ^{13}C NMR spectra related to standards such as tetra-methyl silane (Table 3.1).

The ^{13}C NMR allows analyzing the carbon backbone of organic components for defined organic molecules (Breitmaier and Voelter, 1989). For

Table 3.1 Chemical shift for OM functional groups depending on the position in the molecule and the type of signal in coupled ^{13}C NMR spectra

Functional group	Position in the molecule	Chemical shift (ppm)	Type of signal
Alkyl-C	$\text{H}_3\text{C}-\text{C}$	0–27	Quartet
	$\text{C}-\text{CH}_2-\text{C}$	15–45	Triplet
	$\text{C}-\text{CH}-\text{C}$	20–70	Duplet
O-alkyl	$\text{H}_3\text{C}-\text{O}$	40–60	Quartet
	$\text{C}-\text{CH}_2-\text{O}$	40–80	Triplet
	$\text{C}-\text{CH}-\text{O}$	48–87	Duplet
N-alkyl	$\text{H}_3\text{C}-\text{N}$	20–55	Quartet
	$\text{C}-\text{CH}_2-\text{N}$	36–70	Triplet
	$\text{C}-\text{CH}-\text{N}$	45–75	Duplet
Alkene	$\text{C}=\text{C}$	90–155	Single
Aromatic	$\text{C}-\text{H}$	92–165	Duplet
Carboxylic acid or ester	$\text{H}/\text{R}_3\text{CO} \diagdown \text{C}=\text{O}$ $\text{R}_3\text{C} \diagup$	150–180	Single
Amides	$\text{R}_2\text{N} \diagdown \text{C}=\text{O}$ $\text{C} \diagup$	160–185	Single
Aldehydes or ketones	$\text{H}/\text{C} \diagdown \text{C}=\text{O}$ $\text{C} \diagup$	170–205 or 175–225	Duplet or single

Bold phase characters indicate the carbon atoms that cause the chemical shift.

Adapted from [Hesse et al. \(2005\)](#).

illustration, the ^{13}C NMR spectrum of 1-nitro-propylene ([Fig. 3.5](#)) has signals at 10–20 ppm that are specific for alky groups, and signals at 130–140 ppm are typical for $\text{C}=\text{C}$ groups (the signals at 78 ppm are from the solvent: deuteriochloroform, and the signal at 0 ppm is from tetra-methyl silane that is used as a reference).

The decoupled ^{13}C NMR example ([Fig. 3.5A](#)) indicates that the OM molecule consists of C atoms in three different functional groups: one signal is from an alkyl-C (i) (13 ppm), while two C atoms (ii) and (iii) are within relative similar chemical environments (i.e., two different C atoms in $\text{C}=\text{C}$ group: one is combined with the nitro group and the second is combined

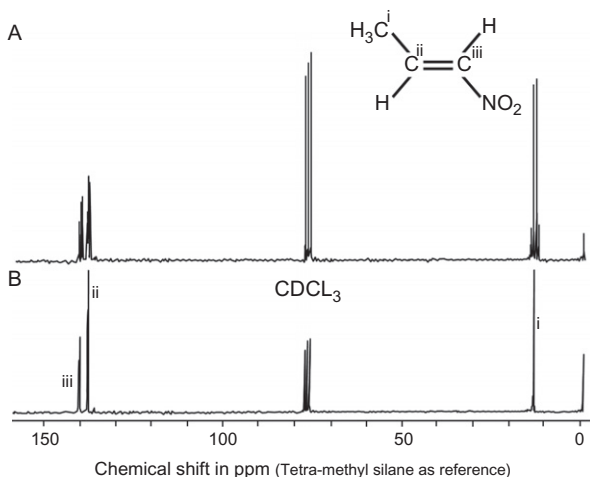


Figure 3.5 (A) Coupled and (B) decoupled ^{13}C NMR spectra (50.3 MHz) of 1-nitro-propylene dissolved in deuteriochloroform (CDCl_3), measured in 1990 at the Institute of Organic Chemistry, University of Essen with a XL200 NMR spectrometer (Varian). The (i), (ii), and (iii) indicate the marked C atoms in the molecular structure of 1-nitro-propylene, and CDCl_3 marks the peaks for C atoms in the deuteriochloroform.

with the CH_3 group) that create relative similar chemical shifts (141 and 138 ppm, respectively). For obtaining more information, the spectrum could be used in the coupled mode resulting in a splitting of the signals. Splitting provides information on the combination between C and H atoms (Fig. 3.5B) indicating that the molecule contains one alkyl- CH_3 , one CH within the $\text{C}=\text{C}$ groups that is bond to a NO_2 group, and one to a CH_3 group. Based on such information and the elemental composition ($\text{C}_3\text{H}_5\text{NO}_2$), the structure of the 1-nitro-propylene molecule could be determined as $\text{CH}_3-\text{CH}=\text{CH}-\text{NO}_2$ with ^{13}C NMR.

The signal intensity in NMR spectra is affected by (i) the relaxation time, which is proportional to differences in occupation of different energetic levels, (ii) measurement conditions, and (iii) on the nuclear Overhauser effect (NOE): The NOE causes much higher intensities for H-containing C atoms (Friebolin, 1999) as compared to quaternary C atoms (e.g., COOH), which may result in two times higher intensities especially for CH_3 , and CH_2 groups. The ^{13}C NMR signal intensity decreases in the sequence $\text{CH}_3 > \text{CH}_2 > \text{CH}$ in Fig. 3.5. A quantitative interpretation of ^{13}C NMR is possible if the following technical measures are considered (Friebolin, 1999): high pulse power and low signal width, high digital

resolution of the peaks, adequate pulse frequency, and suppression of the NOE (e.g., by addition of paramagnetic elements). However, such measures result in long measurement time of days to weeks for a single sample. Note that the ^{13}C isotopes represent only about 1% of the whole C in a soil sample. The low amount of ^{13}C isotopes in nature limits, especially in case of solid-state ^{13}C NMR analysis, the spectral resolution and complicates the interpretation of the NMR spectra (Breitmaier and Voelter, 1989).

4.3. IR/FTIR spectroscopy

IR and FTIR spectroscopy uses light from 2.5 to 25 μm wavelength. In IR spectroscopy, the light is split into two identical beams running in parallel; one is sent directly through the sample and the measurement chamber and the other is sent only through the chamber to account for the air as the background (Fig. 3.6A). In IR spectroscopy, the light of defined wavelength was sent successively through the sample. In FTIR spectroscopy (e.g., Gottwald and Wachter, 1997), the whole spectrum of IR light is sent directly through the sample in the measurement chamber by using a Michelson interferometer (Fig. 3.6B). The interferometer allows consecutively selecting individual wavelength from the beam. The signals arriving at the detector are collected to obtain an interferogram, which is transformed into an absorption spectrum. The transformation is carried out by using the Fourier transformation and a background correction, accounting for the composition of the air in the measurement chamber. For this correction, a background spectrum has to be measured separately by sending the light beam through the

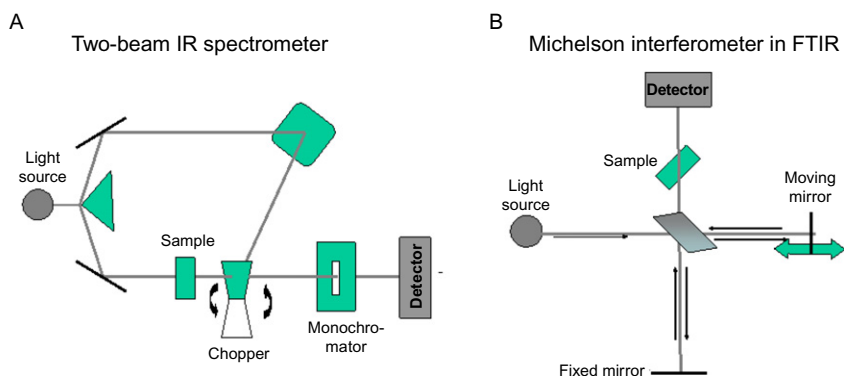


Figure 3.6 Schemes of (A) a two-beam IR spectrometer and (B) a Michelson interferometer within an FTIR spectrometer. Adapted from Hesse et al. (2005).

empty measurement chamber. Note that since the 1980s, improved computer technology offers the possibility to use the Fourier transformation as a standard technique so that the FTIR spectroscopy became the common technique (e.g., Hesse et al., 2005).

The spectra of both techniques, IR and FTIR, are similar and the information obtained is identical. While IR spectroscopy yielded analogous spectra usually measured in transmission mode (i.e., light intensity at the detector), FTIR spectra are usually measured in absorption mode (i.e., light absorbed by the sample as the difference between incoming and detected light intensity), and these spectra can also be converted into transmission spectra. The FTIR spectroscopy has the advantage of an independent background correction, a shorter measurement time, and direct analyses such as baseline correction and spectral subtraction (e.g., Senesi et al., 2001).

The FTIR spectra show characteristic absorption bands that depend on the type of the molecular bonding in OM functional groups (see Table 3.2 for example). For analyzing solid samples such as soils with FTIR spectroscopy, the following sample preparation techniques could be used (e.g., Hesse et al., 2005):

- a. Nujol technique: The sample is mixed with a mineral oil (Nujol) in a thin layer on a flat disk of KBr and covered by a second KBr disk. The sample between the disks is analyzed with FTIR transmission technique (i.e., the IR light is sent through the sample) in absorption mode. Main disadvantage of the Nujol technique with respect to analyzing soil samples is that the overlapping absorption bands of the mineral oil limit the interpretation of the absorption bands of the OM functional groups.
- b. Attenuated total reflection (ATR) technique: A small amount of sample on a sample holder is pressed closely against a plane and polished plate of Zn selenide. The IR light is repeatedly reflected at the polished plates while passing the sample (Fig. 3.7A) and finally detected as an ATR spectrum.

This procedure is useful for finely ground or relatively flexible samples like root samples. The spectra (Fig. 3.8, IIA–C) obtained from differently colored regions at a cutting along a fossil reed root (Fig. 3.8, IIA–C) show differences in the absorption bands. The spectrum of the dark colored root edge (Fig. 3.8, IIA) show smaller absorption intensities of CH and CO bands as compared to the spectrum obtained from a region located in the center of the root (Fig. 3.8, IIC). The combination of microscope with ATR technique allows an *in situ* analysis of the surface of such roots samples. The advantage is that ATR requires only small amounts (i.e., <1 mg) of a sample, no addition of chemicals, and short preparation time.

Table 3.2 Absorption bands characterizing OM functional groups in FTIR, the wave number (WN), and the type of vibration and bonding

Classes of substances	Functional groups	WN (cm ⁻¹)	Type of vibration and type of bonding
Alkyl	CH₃ ; CH₂ ; CH	2950 to 2850 2860 to 2800	Asymmetric CH stretching Symmetric C—H stretching
Polysaccharides	Alkyl-O-alkyl (C—O—C)	1150 to 1100	C—O stretching
Carboxylic acids	$\begin{array}{c} \text{HO} \\ \diagdown \\ \text{C}=\text{O} \\ \diagup \\ \text{C} \end{array}$	1740 to 1698	C=O stretching
Esters, aldehydes, ketones, amides	$\begin{array}{c} \text{RO} \\ \diagdown \\ \text{C}=\text{O} \\ \diagup \\ \text{C} \end{array}$ $\begin{array}{c} \text{R} \\ \diagdown \\ \text{C}=\text{O} \\ \diagup \\ \text{C} \end{array}$		R: alkyl chains for esters R: H for aldehydes R: alkyl chains for ketones R: amino group for amides
Anionic carboxylic acid groups and carboxylic acid groups conjugated with C=C groups	$\begin{array}{c} \text{O}^- \\ \diagdown \\ \text{C}=\text{O} \\ \diagup \\ \text{C} \end{array}$ $\begin{array}{c} \text{C}=\text{C} \\ \diagdown \quad \diagup \\ \text{C} \quad \text{C}=\text{O} \\ \diagup \\ \text{C} \end{array}$	1620 to 1680	C=O stretching
Water, hydroxide, phenols, alcohols		3750 to 2700	OH stretching
Water, hydroxide, phenols, alcohols	O—H; hydrogen bonding	1660 to 1600	OH bonding (i.e., in hydration layers of soil phyllosilicates; Mendelovici et al., 1995; Spaccini et al., 2001)

Bold phase characters indicate the carbon atoms that cause the chemical shift.

Adapted from Hesse et al. (2005).

For highly resolved ATR spectra, however, close contact between the Zn selenide plates and the sample is needed to exclude air. Thus, for inelastic soil samples with particles >0.02 mm, the high pressure needed for the establishing contact may destroy the Zn selenide plates (Kubicki et al., 1999).

- c. Potassium bromide (KBr) technique: The finely ground sample is mixed with KBr (i.e., sample dilution <5%) and both again finely

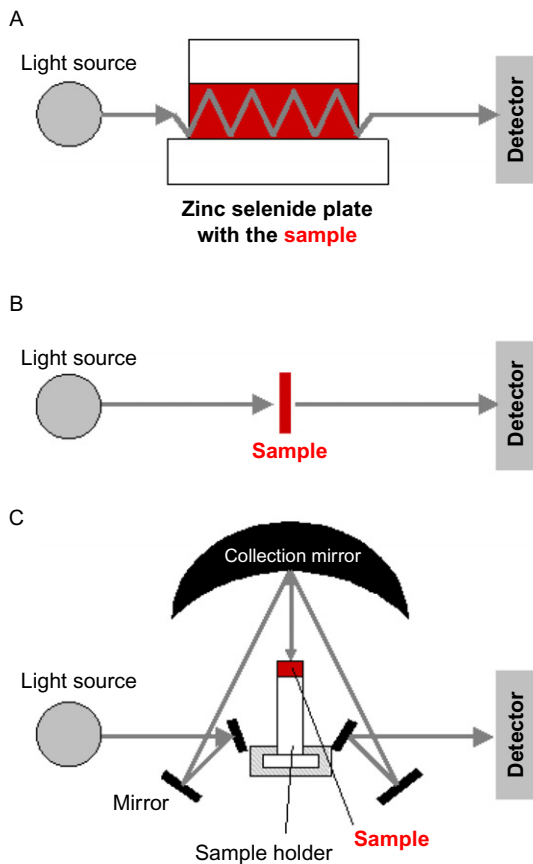


Figure 3.7 Schemes of IR light beam used in (A) attenuated total reflection (ATR), (B) KBr, and (C) DRIFT technique. Adapted from Hesse et al. (2005).

ground in an agate mortar. The mixture is converted into pellets by using a 9-Mg compactor, placed onto the sample holder, and analyzed by FTIR transmission technique in adsorption mode (Celi et al., 1997; Günzler and Böck, 1990) (Fig. 3.7B). The KBr technique yields highly resolved spectra that are easy to interpret; the adsorption bands of the OM functional groups are not affected by those of the KBr. However, disadvantages are that KBr technique requires mixed and finely ground samples and that acidic functional groups in organic molecules will exchange their protons by K^+ which may cause a shift of the $C=O$ absorption bands from the region at 1700 cm^{-1} to that at 1600 cm^{-1} (Günzler and Böck, 1990).

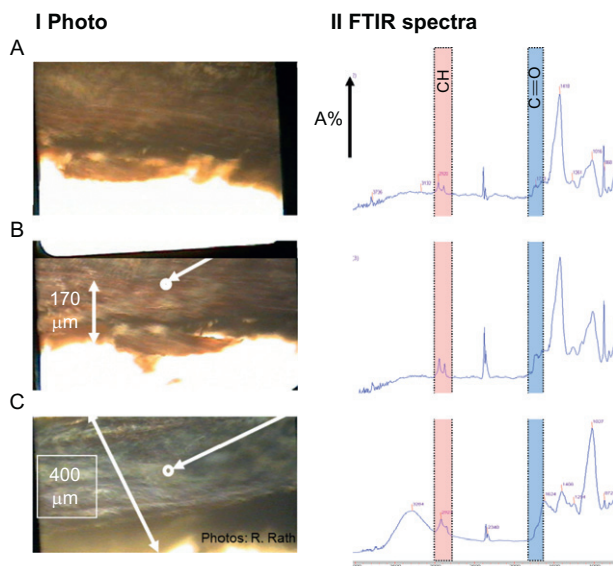


Figure 3.8 Microscopic photos (I) of different regions of a longitudinal cut of a fossil reed root sampled in a fen. Location (A) describes a darker region near the outside of the root, (B) is a region located about 170 μm inside, and (C) a region in the central part of the root. The ATR technique analyzed a region of about 20 μm in diameter, marked by a white circle. The ATR spectra (II) of the regions marked by white circles are shown at the right-hand side.

- d. Combination of FTIR and a microscope: FTIR microscopy (Aparicio et al., 2002; Gregoriou and Rodman, 2002) used intact samples embedded in polyresins and thin sections allowing a spatially resolved analysis of surfaces (Mertig et al., 2002) and a spatial allocation of functional characteristics of organic components to micromorphological structures (Aparicio et al., 2002; He et al., 2002; Zieba-Palus, 1997). However, the FTIR signals of the resins may overlap with those of the OM in the sample (Arocena et al., 1995). Furthermore, the resin may interact with mineral soil components (i.e., C=O-functional groups with Ca^{2+} - or Fe^{3+} -ions) causing shifts in absorption bands that complicate the interpretation of the spectra (Aparicio et al., 2002; Dupuy and Douay, 2001). An alternative is offered by the DRIFT technique.
- e. Diffusive reflectance infrared (DRIFT) technique: For DRIFT analysis, ground bulk or diluted dry soil samples are typically filled in small cups (e.g., Gottwald and Wachter, 1997). A dilution in KBr is required for analyzing OM fractions, while for bulk samples with OM contents

below about 10%, a dilution is not required (Reeves, 2003). The IR light is sent onto the plane and homogeneously prepared sample surface, and the diffusively reflected light is collected (Fig. 3.7C) and DRIFT spectra are displayed in reflectance mode. The DRIFT technique has been used as a relatively rapid technique to analyze composition and dynamics of OM in soil (e.g., Fuller and Griffiths, 1978; Niemeyer et al., 1992; Reeves et al., 2001; Spaccini et al., 2001; Tremblay and Gagne, 2002); however, DRIFT yields less highly resolved spectra as compared to the KBr technique (Haberhauer and Gerzabek, 1999). For analyses of soil samples, the advantage is that DRIFT technique is relatively easy; however, the standard procedure is applicable only to finely ground samples. The DRIFT technique combined with a horizontal positioning device could also be used for analyzing samples like plates used for thin layer chromatography (Böcker, 1997). But it has not been used for *in situ* analyses of OM composition at intact soil surfaces probably caused by problems linked to surface properties of undisturbed soil samples (uneven surfaces and contents of coarser particles). Relatively large particles, for instance, may limit the use of Kubelka–Munk (KM) transformation of DRIFT, which was developed for finely ground samples (Kortüm et al., 1963).

4.3.1 Interpretation of FTIR spectra

For characterization of OM composition, the spectra can be used to qualitatively identify OM functional groups and their relative amounts in comparable samples (e.g., CH/C=O ratios; Ellerbrock et al., 1999a, 2005). In addition, FTIR spectra displayed in absorption mode may be quantitatively evaluated to analyze the content of OM functional groups in soil samples after calibration. Such quantification is based on the relation between the height of absorption bands, that is, absorbance, A , (%) and the amount of functional groups, expressed by the law of Lambert–Beer (Eq. 3.3) as (Hesse et al., 2005):

$$A = \varepsilon cd \quad (3.4)$$

with ε , an molar extinction coefficient (mol cm^{-2} , however, in general mentioned without a dimension) that is specific for each component and each wave number, c , the molar concentration of functional group (mol l^{-1}), and d , the thickness of the sample (cm). For example, the height of the absorption band at 1750 cm^{-1} can be used to determine the amount of carboxylic acid groups if ε for the wave number and component as well as the

thickness is known (e.g., [Günzler and Böck, 1990](#)). For the quantification of spectra obtained by DRIFT technique in reflectance mode, the KM transformation has to be carried out such that band heights will be approximately linearly related to the relative amounts of functional groups ([Fuller and Griffiths, 1978](#); [Mitchell et al., 1994](#)). Note that for determining the complete molecular structure of OM, the results from FTIR spectral analysis need to be combined with additional information on the C backbone of molecules that can be obtained by NMR, elemental analysis, and mass spectroscopy.

4.3.2 A note on PCA/PLS analysis for determining soil C_{org} contents

More recently, the DRIFT spectroscopy was combined with principal component analysis (PCA) and partial least square (PLS) analysis for the quantification of SOC contents in neat (nondiluted) ground soil samples to develop a relatively cheap method to analyze SOC contents of a large number of soil samples (e.g., [Reeves, 2003](#)). Note that the PCA/PLS method first establishes a calibrated relation between spectral data and SOC content by using PLS on a subset of the samples and second validates the relation by using the remaining data set ([Ludwig et al., 2008](#)). Like near-IR spectroscopy (e.g., [Stenberg et al., 2010](#)), the mid-IR spectroscopy (4000 to 400 cm^{-1}) discussed here could be used to assess C_{org} and N contents of soils and litter ([Ludwig et al., 2008](#)). However, the mid-IR spectroscopy offers more detailed information on OM composition since the absorption bands of functional groups like $C=O$ and CH are less overlapping in mid-IR as compared to near-IR spectroscopy ([Hesse et al., 2005](#)). Details on PCA/PLS procedures are beyond the scope of this chapter since it contributes only little information to the analysis of SOM composition.



5. APPLYING FTIR SPECTROSCOPY IN ORGANIC CHEMISTRY AND IN SOIL SCIENCE

The above-mentioned spectroscopic techniques are generally used in chemistry as complementary methods to analyze the composition of defined OM in terms of the molecular structure. The requirement for analyzing mixtures (e.g., water and oil) for the application of the spectroscopic techniques in chemistry is that spectral signals are known and easily be distinguished, also the mixtures need to consist of maximal two or three well-known components, or the components of the mixture belong to the same substances class. The spectral methods were mainly used to detect impurities

(e.g., spots within plastic foils; Böcker, 1997) or to detect and quantify a certain component with a well-known composition within a relatively homogeneous mixture that show easily detectable signals (e.g., glucose in blood; Shen et al., 2003), or CO₂ in gases (IR detector in elemental analysis; DIN ISO 10694, 1994).

5.1. Spectral analysis of defined organic components and SOM

In contrast to defined OM substances (e.g., quinine, glucose, ascorbic acid) or substance classes (e.g., fatty acids), OM in soil and fractions is not a well-defined substance but a complex mixture of organic components (Hayes and Clapp, 2001). SOM is characterized by differing compositions, microbial degradability, and turnover time (von Lützow et al., 2006). The NMR and FTIR spectra of mixtures like SOM will give information on the *mean* functional composition. OM structures as suggested by Stevenson (Fig. 3.1) can only illustrate the mean composition of SOM.

The FTIR spectrum of glucose shows a broad absorption band at 3700–3600 cm⁻¹ for the OH groups and a strong sharp absorption band at 1050 cm⁻¹ for the C—O—C groups. The C—H groups—a relative small content in glucose (Fig. 3.9A)—show a hardly detectable absorption band at 2900 to 2800 cm⁻¹ (pink region). Note the absorption bands of the CH groups are in general much lower than those of OH or C—O—C groups (Hesse et al., 2005).

The FTIR spectrum of glycerin (Fig. 3.9C) shows a broad and intense absorption band of the OH band and absorption bands resulting from CH groups that appear as a weak shoulder on the broad OH band, while an absorption band of a C—O—C group is missing because such groups are not part of the glycerin molecular structure. For ascorbic acid, the absorption band of the OH groups shows a fine structure due to the presence of OH groups within different molecular environments (Fig. 3.9D). Note that in case the samples contain an increasing amount of water molecules, such a fine structure will disappear (Günzler and Böck, 1990) caused by increasing inter- and intramolecular interactions via hydrogen bridging, resulting in a high variety of different molecular environments which will strongly overlap and result at the very end in one broad absorption band. In case of the dry samples, such interactions will decrease because most of the OH groups are not within a distance to form hydrogen bridges (Günzler and Böck, 1990).

In contrast to the FTIR of glucose and glycerin (Fig. 3.8A and C), the FTIR spectra of ascorbic acid (Fig. 3.9D) show an absorption band at 1750 and 1650 cm⁻¹ due to C=O groups in the molecular structure (which is not

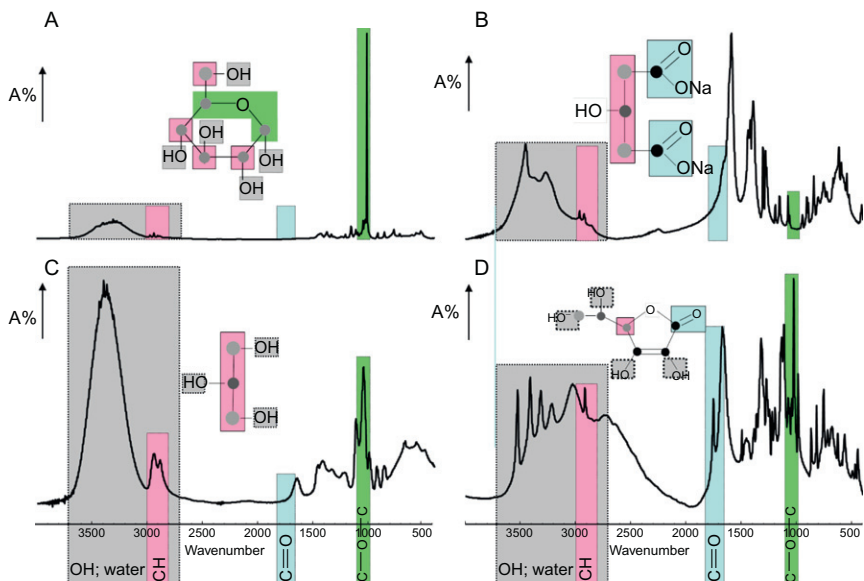


Figure 3.9 FTIR spectra of (A) glucose, (B) sodium citrate, (C) glycerin, and (D) ascorbic acid obtained with KBr technique displayed in absorption mode (A%). The marked areas indicate the absorption bands of CH—groups (pink), C=O—groups (turquoise), C—O—C—groups (green), and OH—groups (gray). The molecular structures are displayed for each substance.

part of cellulose or glycerin). The absorption band at 1750 and 1650 cm^{-1} indicates that ascorbic acid contains beside OH, CH groups and C=O groups as well. In general, the C=O groups that are part of carboxylic acid groups will show absorption bands at 1800 to 1700 cm^{-1} . However, in case the sample was mixed with KBr, a part of the protons within the carboxylic acid groups will be exchanged by K^+ and the FTIR will show the absorption bands of the deprotonated acid group at 1670 to 1640 cm^{-1} (Günzler and Böck, 1990) which is also found in FTIR spectra of sodium citrate (Fig. 3.9B).

For clay mineral components, high-resolved spectra could be obtained as well (Fig. 3.10A): The FTIR spectra of clay minerals show a high absorption band at 1100 cm^{-1} caused by Si—O—Si groups. Such absorption bands strongly overlap with those of the C—O—C groups in organic components. Additionally, the clay minerals show broad absorption bands at 3800 to 2700 cm^{-1} due to OH groups (Fig. 3.10A). In case a clay sample (e.g., kaoline) is mixed with ascorbic acid, the FTIR spectra of the mixture show absorption bands of both components (Fig. 3.10B), whereby the

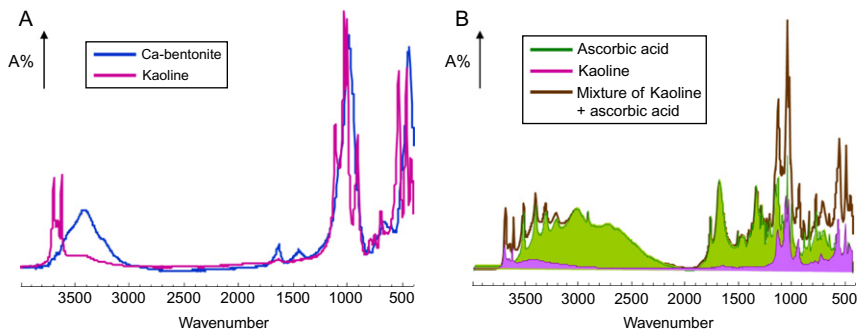


Figure 3.10 FTIR spectra of (A) Ca-bentonite and kaoline and (B) ascorbic acid (green area), kaoline (pink area), and a mixture of both (brown line) measured with KBr technique in absorption mode (A%).

intensity of the absorption bands behaves additive. The brown line in Fig. 3.10B results from the mixture and shows absorption band of both the ascorbic acid and the kaolin sample.

While the green-colored area in Fig. 3.10B is from the FTIR of ascorbic acid and the red area is from FTIR of kaoline, the brown line is the spectrum of the mixture. By using spectral subtraction procedures, the spectra of the single components can be obtained provided that at least one component of the mixture is known (Senesi et al., 2001) and the mixing did not induce any chemical reactions between the components.

FTIR spectra of mixtures (i.e., soil samples) are less resolved (Fig. 3.11) as compared to those of defined organic components like ascorbic acid (Fig. 3.9). Especially, the fine structure in FTIR spectra, often called fingerprint region (i.e., 900 to 400 cm^{-1}), is less resolved in mixtures such as soil samples (Fig. 3.11). The fingerprint region in FTIR reflects structural information on the molecular structure (i.e., status of substitution or conformation) of defined organic components as, for instance, ascorbic acid (Fig. 3.9).

The spectral data will reflect information on the mean composition of SOM or OM fractions. The examples in Fig. 3.11 show that most intense absorption band of soil samples is at a wave number of about 1100 cm^{-1} caused by Si—O—Si groups within quartz and clay minerals (Ellerbrock et al., 1999a; Gottwald and Wachter, 1997). The second most intense absorption bands can be found in the region 3800 to 2700 cm^{-1} . These bands reflect the presence of OH groups (Gottwald and Wachter, 1997) in either OM components (i.e., phenols, alcohols, carboxylic acids) or soil minerals (i.e., in clay minerals or iron or manganese hydroxides among others).

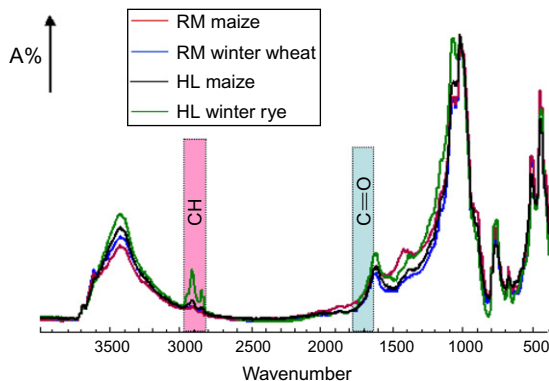


Figure 3.11 FTIR spectra of soil samples from NPK-fertilized plots at long-term field experiments located at Halle and Rotthalmünster measured with KBr technique and displayed in absorption mode (A%); more information on Halle and Rotthalmünster soils can be found in [Ellerbrock and Kaiser \(2005\)](#).

5.2. Applications of FTIR spectral analysis to characterize OM fractions

Although the determination of specific molecular structures of SOM is not possible, FTIR can be used to determine the relative contents of functional groups in SOM when focusing only on those absorption bands that are indicative for OM. [Van der Marel and Beutelspacher \(1976\)](#) found that IR spectra of humic acids from various origins (i.e., peat, coal, lignite, lignin from straw or Black Earth, Podzol, and Chernozem soils) were similar but not identical. These authors assumed that variations in IR were caused by slight differences in the composition of the samples. [Hempfling et al. \(1987\)](#) found by using FTIR differences in the OM composition of forest soils from different sites. They stated that during decomposition of litter, humification, and podzolization, the content of cellulose and lignin decreased, whereas hemicellulose and protein contents appeared unchanged. Land use, crop rotation, fertilization ([Nardi et al., 2004](#)), land use history ([Antil et al., 2005](#)), and soil genesis ([Quideau et al., 2001](#)) were found by using FTIR analysis to effect SOM composition as well.

To allow a more detailed spectroscopic analysis, various OM fractions have been separated from the soil to reduce or eliminate the effects of absorption bands of soil mineral components in FTIR ([Janik et al., 1995](#)) (see also sections 3.1 to 3.4 in this chapter). Effects of different site conditions on the composition of dissolved organic matter (DOM) have been reported

by Guggenberger et al. (1994a,b) for a silty loam soil from loess in a pre-Alpine region. For a clay loam, Gerzabek et al. (1997) found characteristic differences in the composition of fulvic and humic acids only for peat-amended soil, using FTIR and not for other treatments. However, Gerzabek et al. (1997) used a variation of the Nujol technique: They applied the extracts onto disposable IR cards (3 M; polyethylene), dried them, and FTIR analyzed as described for Nujol technique.

The use of OM fractions allowed to demonstrate with FTIR (KBr technique), for example, that the content of polysaccharides in DOM fractions from forest soils changed during the decomposition process (Kalbitz et al., 2003). In another example, the addition of Al to fractions of the DOM was found by FTIR analysis to cause the DOM composition in the precipitated fraction to change as compared to the DOM in solution (Scheel et al., 2008).

For agricultural soils, amount and type of OM functional groups were found to be affected by land use and management practices (e.g., Ellerbrock et al., 1999a,b, 2001a; Gerzabek et al., 1997, 2001; Kaiser et al., 2011; Quideau et al., 2001). FTIR studies on OM composition of SOM fractions from soils with the same genesis indicated effects of farm yard manure (FYM) application such as an increase in carboxyl group content (Ellerbrock et al., 1999b). Antil et al. (2005) found the same effect but for bulk soil and silt-sized fractions. Soils of LTFE (i.e., at least 30 years) listed in Table 3.3 have been used for studying changes in OM composition (e.g., Antil et al., 2005; Ellerbrock et al., 1999a,b; Gerzabek et al., 1997, 2006; Kaiser et al., 2007; Tatzber et al., 2007).

For a comparison of the FTIR spectra of the NaPY-soluble OM fraction of both loamy and sandy soil samples (Fig. 3.12), the intensity of the absorption bands at 1710 and 1690 cm^{-1} (i.e., indicating C—O groups) was highest for the plots that received FYM (Ellerbrock et al., 1999b). According to the high absorption intensities at 1710 and 1690 cm^{-1} , Ellerbrock et al. (1999b) expected relatively high CEC values for the fraction of the soil from the FYM-fertilized plots because the intensity of these absorption bands was reported to be correlated with CEC (Celi et al., 1997; Günzler and Böck, 1990). The CEC of the NaPYseq-soluble OM fraction was found between 25 and 375 $\text{cmol}_c \text{ kg}^{-1}$, which represents a fraction between 1% and 11% of the CEC of the bulk soil (Kaiser et al., 2008).

The CH groups reflect the hydrophobic properties of OM (Morrison and Boyd, 1983) that may be related to the wettability (Capriel, 1997; Capriel et al., 1995); the relative amount of CH groups could be analyzed

Table 3.3 FTIR studies on OM composition of soils from long-term field experiments (LTFE) and applied FTIR techniques

Location	Country	Established in	Texture	References on LTFE	References on FTIR spectra	OM fractions	FTIR technique
Müncheberg	Germany	1963	Sandy loam	Körschens and Müller (1996)	Ellerbrock et al. (1999a,b)	NaPY soluble	KBr
Dedelow		1976	Sandy loam	Körschens and Müller (1996)	Ellerbrock et al. (1999b)	NaPY soluble	KBr
Groß Kreutz		1967	Sandy loam	Körschens and Müller (1996)		NaPY soluble	KBr
Halle		1887	Loamy sand	Merbach et al. (1999)	Kaiser et al. (2007)	NaPYseq soluble, water soluble	KBr
Bad Lauchstädt		1902	Silty loam	Körschens and Müller (1996)	Kaiser et al. (2007)	NaPYseq soluble, water soluble	KBr
Rotthalmünster		1969	Loamy silt		Ellerbrock and Kaiser (2005)	NaPYseq soluble, water soluble	KBr
Broadbalk	UK	1843	Loamy silt			NaPY soluble	KBr
Ultuna	Sweden	1956	Silty clay (37% clay; 41% silt)	Kirchmann et al. (1994)	Gerzabek et al. (1997)	NaPY soluble	Nujol-like technique
					Gerzabek et al. (2006)	Bulk soil, silt-size fraction	KBr
					Tatzber et al. (2007)	Soluble OM fractions	KBr

Prague- Ruzyne	Czech Republic	1955	Clay loam	Skarda and Jokesova (1982)	Filip and Kubát (2001) , Simon (2007)	Humic acid, bulk soil	KBr
Gumpenstein	Austria	1962	Silt loam		Antil et al. (2005)	Particle-size fractions	KBr
Lower Austria	Austria		Loamy silt	Spiegel et al. (2002)	Tatzber et al. (2006)	Bulk soil, humic acid	KBr
Several sites	Brazil	–			Madari et al. (2006)	Bulk soil, particle- size fraction	DRIFT for C content

The analyses were carried out for SOM (bulk soil), particle-size fractions, water soluble, sodium pyrophosphate soluble (NaPY), and sequentially separated sodium pyrophosphate-soluble (NaPYseq) OM fractions.

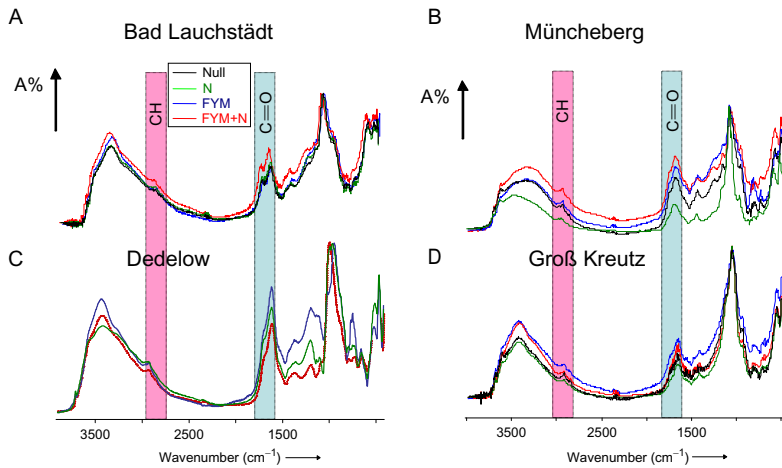


Figure 3.12 FTIR spectra of the sodium pyrophosphate-soluble OM fraction from soils of differently fertilized plots (Null, no fertilizer; N, mineral nitrogen; FYM, farm yard manure; FYM+N, farm yard manure plus nitrogen) of long-term field experiments located at (A) Bad Lauchstädt, (B) Müncheberg (Ellerbrock et al., 1999b), (C) Dedelow (Ellerbrock et al., 2001a,b), and (D) Groß Kreutz. The spectra were measured with KBr technique and displayed in absorption mode (A%) have been corrected with the spectra of the mineral phase (Ellerbrock and Gerke, 2004).

from the intensity of the absorption bands at 2800 and 2950 cm^{-1} . For example, the bands in the spectra representing CH groups are higher for samples from sandy (Fig. 3.12D) as compared to those of a loamy soil (Fig. 3.12A) from plots with similar FYM application (Ellerbrock et al., 2001a,b).

However, the information on the composition of SOM fractions are limited due to artifacts caused by separation procedures (e.g., Hayes, 1985), especially in case sodium hydroxide solution is used (Kononova, 1958), and speciation caused by the solubility of SOM according interactions with mineral surfaces (Piccolo et al., 1993) depending on type of cations (e.g., Ca^{2+} , Al^{x+} , Mn^{x+} , or Fe^{x+}) and amount of clay (Mikutta et al., 2005b) during extraction.



6. RELATIONS BETWEEN SOM COMPOSITION AND SOIL PROPERTIES

It is well known that content and composition of clay minerals and SOM determine the physicochemical nature of soil (Stevenson, 1994) including wettability (e.g., Dekker and Ritsema, 1994) and CEC

(Sposito, 2008). Thus, differences in SOM composition affect soil properties in a similar way as differences in clay mineral composition (e.g., Gerzabek et al., 2001). SOM content and composition are especially important for sandy soils with low clay mineral content, where SOM acts as the major sorption partner for cations (e.g., Almendros, 1995; Hepper et al., 2006). Determination of SOM composition can therefore improve predictions on changes in soil properties as affected by long-term soil management or changes in land use.

6.1. CEC and SOM composition

The CEC of soils characterizes sorption properties that are important, for instance, for describing nutrient retention and supply or the fate of heavy metals and pesticides (e.g., Leinweber et al., 1993). In arable top soils of temperate regions, the CEC is determined mainly by negatively charged sites of (i) 2:1 clay minerals with either permanent (e.g., smectite; Mendonça and Rowell, 1996) or variable charge (e.g., metal oxides; Oorts et al., 2003; Parfitt et al., 1995) and (ii) dissociated carboxyl groups of humic acids (Coles and Yong, 2006; Martyniuk and Więckowska, 2003). The abundance of negatively charged sites depends on (i) the mineral composition of the parent rock and sediments, (ii) mineral weathering processes (Hepper et al., 2006; Oorts et al., 2003), and (iii) pH-dependent structure of SOM (Asadu et al., 1997; Coles and Yong, 2006). The formation of organo-mineral complexes (OMC) may modify the CEC of SOM and clay minerals in soils (Oades, 1989) as follows: The CEC of clays coated with OM can differ significantly from that of uncoated clay minerals (Dudas and Pawluk, 1970; Kahle et al., 2002). Thompson et al. (1989) found the CEC of OMC to be mostly lower than the sum of the CEC values from both organic and inorganic components. The CEC values of soils vary widely depending on the amount and composition of clay minerals and SOM (Kyzioł et al., 2006; Parfitt et al., 1995). The relative importance of SOM for the CEC of the soil increases with decreasing amounts of negatively charged 2:1 clays (Hepper et al., 2006). The CEC of SOM is mainly related to the content of carboxyl groups (Barančíková et al., 1997; Coles and Yong, 2006), which can be quantified with FTIR spectroscopy (Celi et al., 1997).

In previous studies, CEC of OMC was analyzed (e.g., Hepper et al., 2006; Leinweber et al., 1993; Oorts et al., 2003). Such complexes were separated from soil using aggregate and particle-size fractionation techniques including density separation, sieving, and ultrasonic treatment. However,

the CEC of such OMC reflects the CEC of both the OM and the minerals. For analyzing CEC of OM fractions, those fractions need to be separated themselves (i.e., without complexation with minerals). A possibility was offered by studies that determine sorption properties of OM fractions like humic acids from the content of carboxyl group analyzed with FTIR (Barančínková et al., 1997; Celi et al., 1997; Coles and Yong, 2006), among other techniques. Still, the knowledge on the relation between CEC and OM composition is limited, partly because SOM separation techniques are not suitable. For loamy and sandy soils from two LTFE, between 13% and 34% of SOC could be extracted sequentially with NaPY. The composition of this OM fraction (i.e., the content of carboxyl groups) was used to identify long-term effects of crop rotation and fertilizer application by using FTIR (Kaiser et al., 2007).

For the soils from the same LTFE, the contribution of NaPYseq-soluble OM fraction to CEC(soil) was determined by comparing the chemical composition of SOM with that of the OM fraction. The experiments at Halle (HL) and at Bad Lauchstädt (BL) were selected because of their similar age (i.e., about 100 years) and soil pedogenesis (i.e., loam derived from loess). The CEC of NaPYseq-soluble OM was determined by using a slightly modified procedure as described by Leinweber et al. (1993). To simulate an SOM content of 3% in soil (which is nearly 1.5% SOC), Kaiser et al. (2008) used a mixture of 0.03 g of freeze-dried NaPYseq-soluble OM fraction and 4.97 g of acid-purified sand (Seesand, Carl Roth, Karlsruhe, Germany): This mixture was added to a 50-mL glass beaker and stirred with a glass stick for about 2 min until the mixture was uniformly colored. This mixture was then placed in a percolation tube and the CEC was determined according to DIN 19684 (1977). The Ba^{2+} concentration in the combined percolates was determined using an atomic absorption spectrometer (1100B, PerkinElmer, Waltham, MA) to yield the CEC of the NaPYseq-soluble OM in terms of charge equivalents (Kaiser et al., 2008).

The CEC of the NaPYseq-soluble OM fraction per unit mass of soil was found to depend on amount and composition of this OM fraction. The relation between the intensity ratio of absorption bands of the CO and COC groups (CO/COC) for the *per unit mass of OM* and the CEC of the NaPYseq-soluble OM fraction was found to be relatively linear even including the soils of the FYM plots (Fig. 3.13). A similar relation between CO/COC ratios of an NaPYseq-soluble OM fraction and the CEC(PY) has been reported by Kaiser et al. (2008), although sequentially extracted and soils had

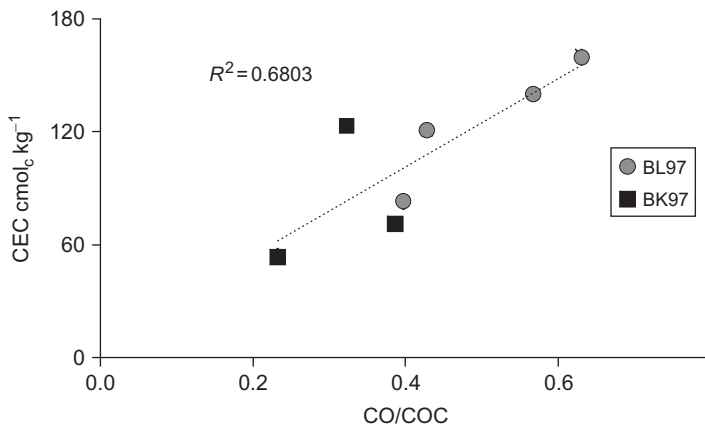


Figure 3.13 Cation exchange capacity (CEC) per unit mass of OM versus the ratio of the intensity of absorption bands of the C=O and the C—O—C groups in FTIR (CO/COC ratio) for the NaPYseq-soluble OM fractions from soil samples of long-term field experiments at Bad Lauchstädt (BL) and Broadbalk (BK). *Modified from Ellerbrock et al. (2001a,b)*

higher clay contents as those reported in [Ellerbrock et al. \(2001a,b\)](#). The linear relationship of the form

$$\text{CEC} = 159.3 \text{CO/COC} \quad (3.5)$$

determined by [Kaiser et al. \(2008\)](#) indicates that for the NaPYseq-soluble OM, the carboxyl group content is similarly related to the CEC, as theoretically expected for pure organic compounds ([Günzler and Böck, 1990](#)).

The relation between the CEC and the relative carboxyl group content (CO/COC ratio) in the NaPYseq-soluble OM ([Kaiser et al., 2008](#)) confirms that the composition determines the sorption properties of OM. However, the limited extractability of the NaPYseq-soluble OM fraction (i.e., also depending on soil mineral and OM composition) still limits transfer of conclusions to the bulk SOM. Note that the extraction procedure may have affected the number of available sorption sites, since the complexation by pyrophosphate removes adsorbed minerals that were originally interacting with the NaPYseq-soluble OM fraction in soil.

6.2. Wettability of soils and SOM composition

Wettability is the ability of a liquid (e.g., water) to spread on a solid surface ([Letey, 1969](#)) and related to the solid–liquid contact angle (CA), which depends on the solid surface free energy and the surface tension of the liquid; the CA is determined at the air–water–solid boundary line ([Letey, 1969](#)).

Values of the CA between 0° and 90° indicate a subcritical (i.e., not hydrophobic) range of water repellence; values $>90^\circ$ indicate a water-repellent porous medium (Morrow, 1976; Siebold et al., 1997).

Attempts to explain soil water repellency focused on the SOM content (e.g., Chenu et al., 2000). These attempts assumed that the SOM provides a large number of hydrophobic components such as aliphatic constituents (Capriel, 1997; DeBano, 1981) and waxes (Franco et al., 2000). In comparison to the SOM content, the OM composition allowed a better prediction of soil water repellence (McKissock et al., 2003). When using the intensity of the absorption bands of the CH groups in FTIR spectra (i.e., wave numbers 2950 and 2800 cm^{-1}), Capriel et al. (1995) estimated a potential wettability and McKissock et al. (2003) tried to relate spectral information to the wettability determined with water drop penetration time tests. However, conflicting results were found by Mataix-Solera and Doerr (2004) and Horne and McIntosh (2000). Negre et al. (2002) suggested that wettability may be better explained by a combined effect of hydrophobic and hydrophilic groups in OM as already stated by Morrison and Boyd (1983) for defined OM. For example, fatty acids, which contain both COOH and CH groups, change from hydrophilic to hydrophobic character if the carbon backbone consists of more than five C atoms (Falbe and Regitz, 1998).

To prove if the ratio between hydrophobic and hydrophilic groups in OM could be used to improve the description of soil wettability, Ellerbrock et al. (2005) determined the ratio by FTIR analysis by using the intensity of the CH bands as a measure for the hydrophobic (A) and the intensity of the CO bands as a measure for the hydrophilic groups (B) and related the obtained CH/CO ratio with the wettability of soil samples. For soils with SOC contents below about 10 g kg^{-1} , the wettability was found to increase with SOC content, while for soils with SOC contents larger than 10 g kg^{-1} , it was found to decrease (Fig. 3.14).

The CA was expected to increase with larger values of the CH/CO ratio since the content of hydrophobic relative to hydrophilic groups increases in that case. When plotting CA versus the CH/CO ratio determined with FTIR for the SOM of bulk soil samples (Fig. 3.14B), the data follow the expected relation except for the samples of the Podzol site. Note that the CH/CO ratio only reflects the mean composition of the OM in the bulk soil, and it did not yield information on interactions between OM and soil minerals or the ratio between OM content and soil mineral surface.

In principle, SOM is adsorbed onto mineral surfaces via PC such as Ca^{2+} , Al^{x+} , Mn^{x+} , or Fe^{x+} . Therefore, both SOM and mineral composition and

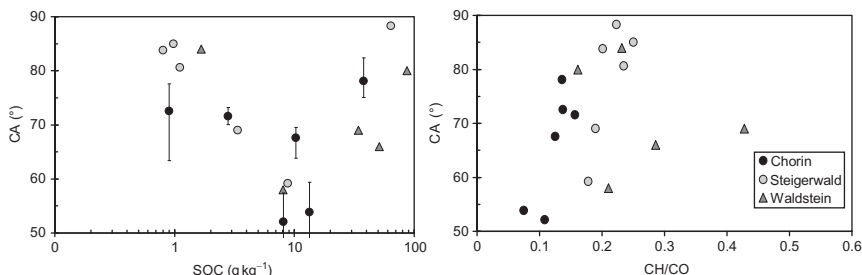


Figure 3.14 Contact angles (CA) related to (A) C_{org} contents and (B) the ratio between hydrophobic C—H and hydrophilic C=O groups (CH/CO ratio) determined by FTIR for bulk forest soil samples from the sites at Chorin, Steigerwald, and Waldstein. Adapted from Ellerbrock et al. (2005).

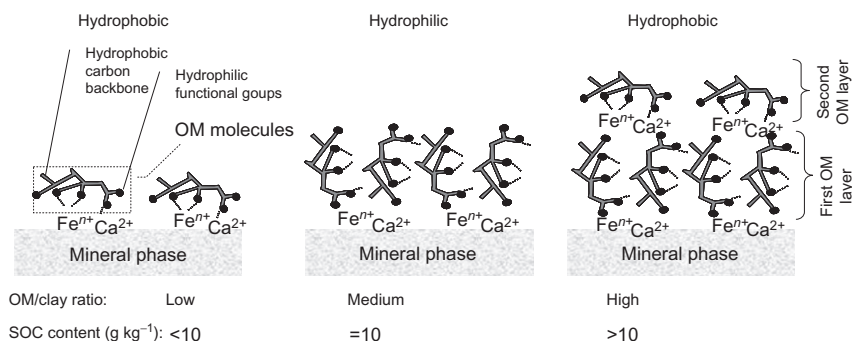


Figure 3.15 Schematic illustration of possible spatial orientations of organic matter (OM), consisting of the carbon backbone chains (solid gray) and the functional groups (solid black circles) toward the mineral surface depending on the relations between the number of OM molecules at the mineral surfaces and the OM/mineral ratio, that is, the ratio between OM and clay content (Ellerbrock et al., 2005).

the presence of PC, may determine the “effectiveness” of SOM to affect soil wettability: In case of a relatively low OM/mineral ratio, all hydrophilic groups of the OM molecules can adsorb to the mineral surface resulting in a sparse flat “monolayer” of SOC (i.e., a single layer of OM molecules) whereby most hydrophobic groups are directed toward the outer surface (Ellerbrock et al., 2005) (Fig. 3.15A). Under such conditions, the hydrophobic groups of SOM are more effective with respect to soil wettability than the mean OM composition (CH/CO ratio) determined by FTIR may suggest.

Depending on the OM/mineral ratio, the spatial orientation of the OM molecules in that layer will change from a sparse flat (i.e., “low,” Fig. 3.15,

left) to a dense upright one (i.e., “medium,” Fig. 3.15, center) (Clapp et al., 2001).

In case of a “medium” OM/mineral ratio, a maximum amount of OM molecules forms a “monolayer” at the mineral surface (Fig. 3.15, center, now in a dense upright orientation) such that, both, hydrophilic and hydrophobic, components will appear at the outer surface. Thus for a “medium” OM/mineral ratio, soil wettability can be expected to correspond with the CH/CO ratios determined by FTIR. In case of larger OM/mineral ratios (Fig. 3.15, right), the outside surface of the “monolayer” can interact with a “surplus” OM molecules if PC are present, while forming a second layer. The outer surface again becomes more hydrophobic initially since the hydrophilic functional groups again tend toward the inside and the hydrophobic ones toward the outside. With further increasing OM content—forming onion-like structures (e.g., Kleber et al., 2007), more hydrophilic groups may again be directed toward the outside. Thus, the effectiveness of OM composition with respect to soil wettability needs to be weighted at least with respect to the OM/mineral ratio. In order to evaluate the effectiveness of CH/CO ratio, Ellerbrock et al. (2005) introduced an optimized, none-dimensional weighing factor as a function of the SOC/clay ratio (g g^{-1}).

In fact, the effectiveness factor improved also strongly the relation between CAs and the weighted CH/CO ratios for bulk soil OM (Fig. 3.16A). The results suggest that such effectiveness factor may well be described with an exponential relation. For describing the relation between wettability and the CH/CO ratio measured for the NaPYseq-soluble OM fractions, site-specific coefficients were required (Ellerbrock et al., 2005). This could be explained by differences in soil genesis, which are manifested in the content of PC within the soil, and in site-specific compositions of inorganic compounds (i.e., clay minerals, Fe, and Mn oxides).



7. FROM THE LAB TO THE FIELD

7.1. Modifications by soil sampling of structured soils

The characterization of SOM and OM fractions using FTIR spectra has in the majority of the before-mentioned studies been carried out with homogenized or otherwise intensively mixed samples collected mostly from the topsoil horizon. Mixing of collected soil samples can be useful where local-scale heterogeneity effects are of minor interest. The results obtained from spectroscopic analysis of mixed samples can be used, for example, for

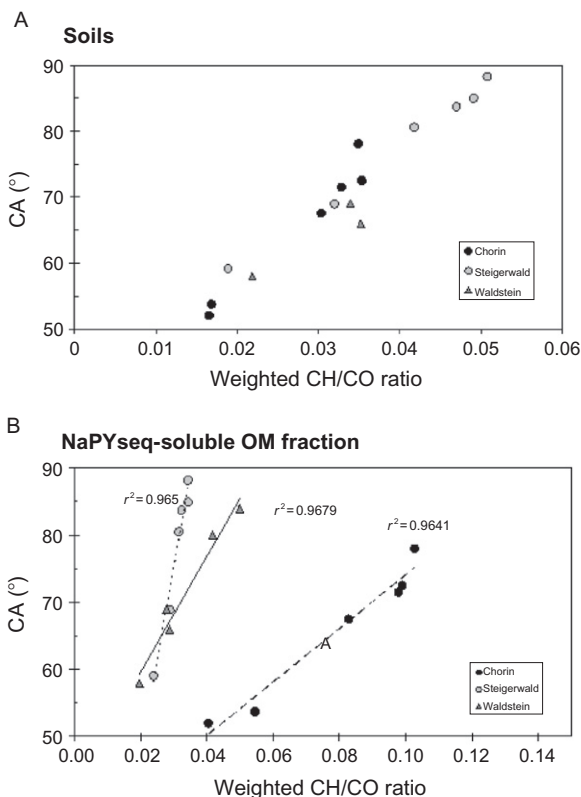


Figure 3.16 Contact angles (CA) related to the weighted ratio between hydrophobic to hydrophilic groups (CH/CO) from FTIR spectra of (A) bulk soil and (B) NaPYseq-soluble OM fractions; samples were from Chorin, Steigerwald, and Waldstein sites (Ellerbrock et al., 2005).

comparing long-term mean OM compositions that develop in an intensively cultivated topsoil of differently managed plots (e.g., Kaiser et al., 2007; Powlson, 1996).

In structured and macroporous soils, however, the local-scale spatial distribution of SOM can be especially important during preferential flow and transport of solutes in the larger and vertically continuous pore space (e.g., see reviews by Köhne et al., 2009a,b). Macropore wall coatings were reported to significantly affect especially the reactive transport in structured soil (Bundt et al., 2001; Eich-Greatorex and Strand, 2006; Mallawatantri et al., 1996). However, when mixing aggregated soil samples, the contribution of the coating will be largely lost (i.e., considering its relatively small mass fraction) and the mixture will mainly reflect the properties of the bulk

soil matrix (e.g., [Bachmann et al., 2008](#); [Horn and Smucker, 2005](#); [Santos et al., 1997](#)).

Recent attempts to describe flow and reactive transport in structured soil (e.g., [Gerke, 2006](#); [Jarvis, 2007](#)) that were based on the determination of local geometry of the macropore space ([Luo et al., 2010](#); [Mooney and Morris, 2008](#); [Ngom et al., 2011](#); [Perret et al., 1999](#); [Tippkötter et al., 2009](#)) and the chemical composition of the pore surfaces (e.g., [Elyeznasni et al., 2012](#); [Kodesova et al., 2008](#); [Köhne et al., 2011](#); [Leue et al., 2010](#); [Rasa et al., 2012](#)) indicate that sorption parameters from mixed samples are not suitable for such analyses in structured soil.

Soil structural surfaces, such as clay-organic coatings on aggregate surfaces, occur, in particular, in illuvial soil horizons such as Bt horizons of Alfisols as a result of particle migration (e.g., [Bronnikova et al., 2000](#)). For biopores, functional domains ([Lavelle, 2002](#)) have been introduced to distinguish the regions close to these pores from the bulk soil matrix, such as “drilosphere” for earthworm burrows (e.g., [Schrader et al., 2007](#)) and “rhizosphere” for the environment of roots (e.g., [Hinsinger et al., 2009](#)). Furthermore, walls of worm burrows can be covered by worm feces or casts (e.g., [Capowiez et al., 2011](#); [Piron et al., 2012](#)), abandoned burrows can be penetrated by plant roots (e.g., [Leue et al., 2010](#)), and the surface of decayed root channels may contain residues of the partly decomposed or remnant roots (e.g., [White and Kirkegaard, 2010](#)) or hydrophobic substances (e.g., [Bengough, 2012](#)). These surfaces can be different in texture, clay mineralogy, and SOM from the bulk soil matrix ([Gerin et al., 2003](#); [Guggenberger et al., 1996](#); [Kaiser and Wilcke, 1996](#)). In contrast to organic coatings of individual particles ([Amelung et al., 2002](#)), a coated surface of preferential flow paths is a porous medium itself. Coatings of soil aggregates or linings along biopore walls ([Ellerbrock et al., 2009](#)) may differ not only with respect to SOM contents but also to the mean OM composition as compared to the soil matrix ([Ellerbrock and Gerke, 2004](#)). The clay mineralogical composition of the coating was spatially variable at the local scale (e.g., [Jongmans et al., 1998](#)).

Since sorption properties are related to OM content and composition of coatings, the information may help characterizing effective parameters in mass exchange terms of two-domain models (e.g., [Gerke, 2006](#)). While the mineral components of aggregate coatings have previously been studied (e.g., [Jongmans et al., 1998](#); [Kaiser and Wilcke, 1996](#)), the quantification of OM composition and physicochemical properties directly at aggregate surfaces is still limited. Previous studies ([Ellerbrock and Gerke, 2004](#)) were still

based on manually removing of a thin layer of coating material about 1 mm from soil aggregate samples and using transmission FTIR with KBr technique. Still, the intact outermost surface (i.e., the interface between matrix and flow path) of soil aggregates or biopores was not analyzed. This interface cannot be separated from the aggregates or biopores in a nondestructive way. Thus, a nondestructive method was required to measure the OM composition at the intact surface. Also, techniques such as density fractionation could not be applied because the original spatial conformation of OM functional groups as in intact soil could not be preserved in density-separated SOM fractions (see Fig. 3.3). To consider the small-scale spatial distribution of SOM in soil at flow path surfaces, a direct analysis at intact surfaces seems to be essential.

7.2. Analyzing soil structural surfaces—Transect and map

A number of spectroscopic methods and techniques are currently available for studying and visualizing the mineral and OM composition of biogeochemical interfaces in soil at the submicrometer scale (Rennert et al., 2012), including X-ray spectromicroscopy, nanoscale secondary ion mass spectrometry, atomic force microscopy, micro-X-ray tomography, and positron emission tomography. Combining the techniques is still a great challenge. However, for the analysis of soil structural surfaces (i.e., surfaces of soil aggregates and biopores), a macroscopic scale characterization of the OM composition requires the information of the properties at the millimeter scale when dealing with preferential flow in macropores.

Depending on the spatial resolution of the instrument, the “effective” properties integrated over the IR beam region were determined using DRIFT spectroscopy in different applications. The local differences in the SOM composition at the surface of soil samples could be determined by stepwise moving the sample along millimeter-scale transects while taking DRIFT spectra (Ellerbrock et al., 2009). These DRIFT transects yielded useful spectra from surfaces of disturbed soil samples as well as from intact aggregates with a relatively smooth and fine-textured surface. Ellerbrock et al. (2009) and Leue et al. (2010; 2011) combined the DRIFT technique with a positioning table to obtain millimeter-scale maps of SOM composition at intact soil structural surfaces. As for the FTIR spectra, the interpretation of these KM-transformed DRIFT spectra (Ellerbrock et al., 2009) focused on the alkyl (CH) and the carboxyl (C=O) absorption bands to characterize hydrophobic and hydrophilic functional groups in OM (Fig. 3.17). The ratio between the intensities of the CH and C=O

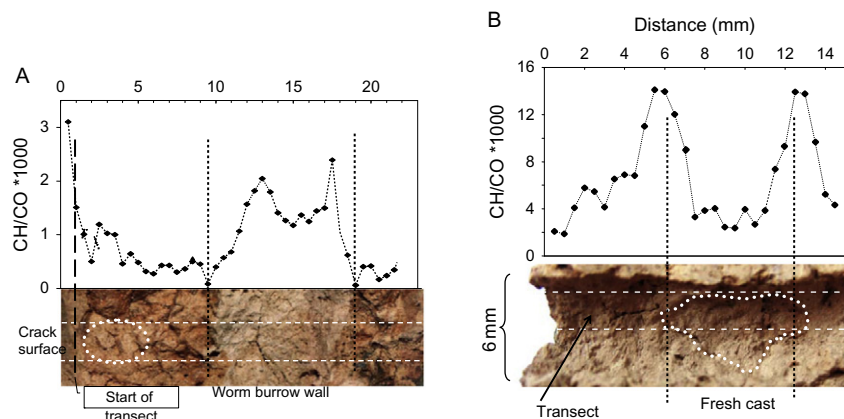


Figure 3.17 Ratios of hydrophobic to hydrophilic groups (CH/CO) ratios measured with DRIFT mapping technique along (A) a 22.5-mm transect at a soil aggregate surface perpendicular to an earthworm burrow and (B) a 14.5-mm transect within an earthworm burrow from an arable subsoil (40–80 cm depth). The vertical black dotted lines (at 9.5 and 19 mm distance) in (A) are used to mark the edges of the worm burrow. *Adapted from Ellerbrock et al. (2009)*

absorption bands (CH/C=O ratio) indicating a potential water repellency of the OM at the surface could then be analyzed in relation to their spatial distribution at the crack surfaces or earthworm burrow walls. By comparing the spectral data with photos of the corresponding sample surfaces, Ellerbrock et al. (2009) attempted a qualitative interpretation of the local distribution of clay or worm casts. The CH/C=O ratios allowed determining the position of root residues and earthworm channels along measurement transects at fine-textured surfaces of the macropores that could potentially act as preferential flow paths.

In order to analyze the two-dimensional local distribution of OM composition at intact aggregate surfaces and earthworm burrows, Leue et al. (2010; 2011) used a DRIFT mapping procedure. Examples of the millimeter-scale maps of OM composition in form of the CH/C=O ratios could be obtained for a number of intact structures from the Bt horizon of a Luvisol (Fig. 3.18). The results of DRIFT mapping indicated that the intact surfaces of aggregate samples could be grouped into regions of earthworm burrows, root channels, and aggregate coatings that were clearly distinguishable (Leue et al., 2010). Among these groups of surfaces, the potential hydrophobicity (i.e., CH/C=O ratio) was mostly higher for worm burrows as compared to aggregate surfaces or root channels (Fig. 3.19).

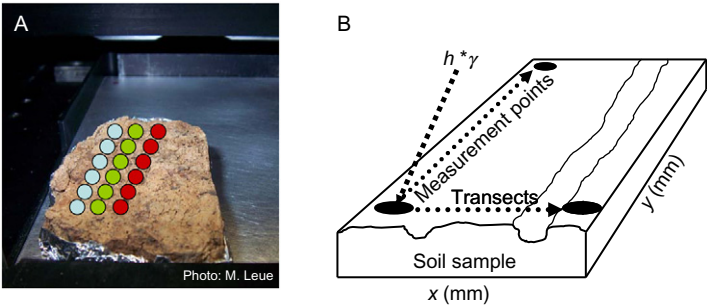


Figure 3.18 Illustration of the DRIFT mapping technique (A) photo of the positioning table with a soil aggregate sample, differently colored circles indicate DRIFT measurement points at the intact aggregate surface and (B) schematic illustration of a measurement transect along a soil structural surface.

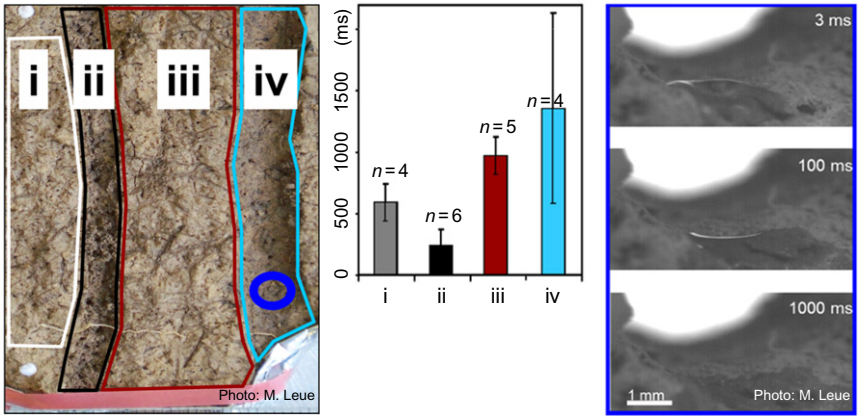


Figure 3.19 (A) Photo with marked regions at a soil structural surface, (B) water droplet infiltration times (mean values and standard deviations) measured at these regions, and (C) photos of the time-dependent contours of a water droplet placed at area *d* (i.e., at the blue circle in the left photo A). The regions indicate the left crack surface (i), an older earthworm burrow (ii), the central crack surface (iii), and a younger earthworm burrow (iv). Adapted from [Leue et al. \(2010\)](#).

Combined water drop infiltration tests ([Leue et al., 2010](#)) indicated that the infiltration was slowest for regions with a high DRIFT-measured potential wettability, such as younger worm burrow surface; infiltration was rapid for uncoated crack surfaces that had a low CH/C=O ratio. For the older worm burrow wall, the infiltration times were relatively short despite a relatively higher CH/C=O ratio ([Fig. 3.19](#)). Of course, water infiltration

depended, in addition to the CH/C=O ratio, also on the pore size and porosity of the burrow wall, which was found to be larger for older as compared to the younger burrow.

The results obtained by DRIFT mapping indicate that coatings at macropore and flow path surfaces differed locally in terms of OM composition, distribution, and possibly also in wettability, indicating yet unknown implications for preferential movement of water and reactive solutes. For more closely linking CH/C=O ratios with water infiltration, additional studies are required to determine (i) a factor that describes the effectiveness of OM composition for wettability depending on the ratio between OM and mineral (Ellerbrock et al., 2005), (ii) a spatial allocation of droplet and DRIFT measurement positions, and (iii) a spatial interpolation of the CH/C=O ratio across the droplet-affected areas. Finally, the analyses of the spectral information need to be refined to allow for a more quantitative interpretation.



8. SUMMARY AND CONCLUSIONS

This overview revealed the importance of characterizing both the content and composition of SOM. The SOM composition could be analyzed by using spectroscopic techniques like NMR and FTIR. Both techniques are in principle complementary and based on the same physicochemical principle, which is the interaction between electromagnetic radiation and submolecular components. The difference in the interactions is caused by the wavelengths of electromagnetic radiation that are used: FTIR spectroscopy is based on light of wavelength between 2.5 and 25 μm that interact with molecular bonds; NMR spectroscopy is based on the interaction between radio waves and the orientation of nuclear magnetic spins of atoms (i.e., ^{13}C and ^1H isotopes) in an external magnetic field. Both FTIR and ^{13}C NMR spectra provide information on SOM composition in terms of functional groups like CH or C=O groups. The NMR spectroscopy can be limited by the presence of paramagnetic elements like iron in soil. In comparison to NMR, the FTIR spectroscopy is a relatively fast technique for analyzing solid samples. FTIR provides information on the composition of organic and mineral soil components.

Effects of climate, soil genesis, land management, and land use history on SOM composition could be analyzed with FTIR spectroscopy. Although developed for finely ground and diluted samples, the DRIFT technique was found applicable to undiluted soil samples when focusing the

interpretation on specific OM absorption bands not overlapped by mineral absorption bands.

FTIR-based OM composition suggested a linear relation between the content of carboxyl groups and the CEC for a sequentially extracted OM fraction. This relation confirmed that the OM composition determines the CEC of SOM and thus also of the bulk soil. Similarly, the ratio between hydrophobic and hydrophilic functional groups determines a potential wettability of SOM and the bulk soil. Land use, SOM composition, and soil properties are, of course, interrelated. Because of this interrelation, changes in OM composition should be considered when predicting transport processes in soil and landscapes.

The transport parameters relevant for structured soils require information of the properties of potential flow path surfaces such as aggregate and biopore surfaces that are in direct contact with the percolating solute. The spatial distribution of OM composition at structural surfaces needs to be considered for adequate transport descriptions. With a DRIFT mapping procedure, millimeter-scale distributions of OM composition at intact structural surfaces could be obtained. At these surfaces, regions of higher and lower potential wettability could be distinguished.

Comparisons of OM composition and water infiltration seem promising for the aim of relating spectral information to soil properties. The mapping technique still requires further developments for considering effects of irregularly shaped surfaces on the IR light scattering. Here, the DRIFT mapping data need to be corrected for the effects of the surface geometry on the spectral intensity. The time-dependent changes in SOM composition and spatial heterogeneity effects are still widely unknown.

The presented techniques using FTIR spectroscopy allow analyzing intact sample surfaces in defined limits (i.e., relatively smooth and fine-textured surfaces). The spectra contain also information on soil mineral components not yet analyzed. Further developments and more quantitative interpretation of spectra may help improving the understanding of small-scale differences in SOM distributions and local-scale properties when studying flow and transport structured soils. The present work may provide the physical and chemical basis for application and interpretation of spectroscopic methods to structured soil.

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